Disentangling Complex Forest Canopies Using MS^{3}Mamba With Hierarchical HSI-LiDAR Fusion for Robust Tree Species Classification

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***Abstract*—Accurate tree species classification is critical for sustainable forest management, but is hindered by the high intraspecies variability and low interspecies separability, inherent in hyperspectral imaging (HSI) data. While the fusion of HSI with light detection and ranging (LiDAR) data offers a promising solution, existing fusion methods exhibit a critical limitation, as they are often designed for general scenes with high interclass variance, and thus are unable to resolve the subtle spectral ambiguities of complex, occluded forest canopies, leading to significant pixel-level misclassifications. To enhance the separability of tree species with high spectral similarity and to mitigate the resulting misclassifications, we propose multisource spectral–spatial Mamba (MS^{3}Mamba), an end-to-end hierarchical fusion architecture specifically engineered for tree species classification. MS^{3}Mamba strengthens feature learning by fusing shallow and deep features extracted by the spectral-structural feature encoding (SSFE) and global canopy context modeling (GCCM) modules, respectively. It synergizes local features from a multiscale convolutional neural network (CNN) with global features from spatial and spectral Mamba blocks for complementary integration, utilizing two independent modal data streams to avoid mutual interference. Furthermore, the tree-aware detail refinement (TADR) module optimizes the spatial–spectral fused features at tree canopy boundaries to enhance classification accuracy. To rigorously validate model robustness, we establish the MNFPTree-HL dataset, a new benchmark from a structurally complex natural secondary forest, comprising co-registered, submeter HSI (0.3 m) and LiDAR (0.1 m) data with an extensive ground-truth inventory (709 trees, 11 species) that addresses the critical research gap for evaluating the performance under these challenging conditions. Comprehensive experiments demonstrate that MS^{3}Mamba achieves state-of-the-art (SOTA) performance by effectively disentangling spectrally similar species and accurately discriminating complex canopy boundaries, particularly on our new MNFPTree-HL benchmark.**

***Index Terms*—Complex forests, fine-grained canopy boundary, Mamba, multimodal fusion, tree species classification.**

NOMENCLATURE

Site ID	Identifier for the study site.
Plot ID	Identifier for the plot within each site.
Tree ID	Identifier for the tree within each plot.
Species	Tree species information.
DBH	Diameter at Breast Height.
H	Tree Height.
CBH	Crown Base Height.
N	Crown extent to the North.
S	Crown extent to the South.
W	Crown extent to the West.
E	Crown extent to the East.
X	Longitude coordinate.
Y	Latitude coordinate.

I. INTRODUCTION

AN ACCURATE spatial information on tree species distribution is crucial for forest inventory, biodiversity assessment, and sustainable management [1]. Hyperspectral imaging (HSI) has emerged as a cornerstone technology for this task due to its high spectral resolution [2]. HSI provides high sensitivity to the subtle spectral variations among tree species, which are driven by their distinct biochemical and biophysical properties [3]. This intrinsic capability gives HSI a distinct advantage in discriminating among spectrally similar species, thereby underpinning the potential for large-scale and fine-grained classification [4]. Consequently, HSI effectively overcomes the limitations of multispectral or RGB, which are often unable to resolve subtle differences, establishing it as a key technological approach for tree species identification [5].

In recent years, studies have focused on fully exploiting the potential of HSI data by advancing model architectures, feature representations, and training strategies. For instance, some studies [4], [6], [7] have enhanced spatial feature extraction by incorporating texture-aware self-attention to mitigate spectral

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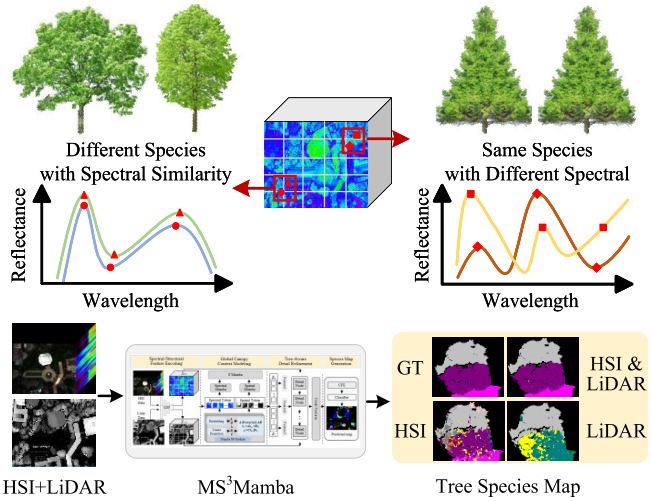


Fig. 1. Challenges in fine-grained tree species classification and the necessity of multisource data fusion. The upper part depicts the inherent challenges of tree species classification using solely HSI data: different tree species with similar spectrum and the same tree species with different spectra. The lower part of the three types of input (HSI only, LiDAR only, and HSI & LiDAR fusion) demonstrates that the fusion of HSI and LiDAR data significantly mitigates these issues, leading to superior tree species classification results compared with a single-modality approach.

variability or by designing learnable morphological operators to better capture complex crown geometries [6]. Concurrently, other works have addressed sample scarcity through systematic data augmentation for few-shot learning [5] or improved computational efficiency by integrating superpixel information into cascaded architectures [7]. Despite these advancements, HSI-based classification was fundamentally constrained by two inherent challenges [8]: intraspecies spectral variability and interspecies spectral similarity. As illustrated in Fig. 1, the same species may exhibit heterogeneous spectra due to varying environmental conditions, while different species may display overlapping spectral features due to similar biochemical or structural properties [3]. For example, the spectra of a single pine species can vary significantly across different slope aspects, whereas the reflectance of pine and fir may converge in certain wavelength ranges. This phenomenon of high intraclass variance and low interclass separability undermines the reliability of any method solely dependent on spectral data. Consequently, this inherent spectral ambiguity limits HSI's effectiveness in operational tree species mapping and necessitates the exploration of complementary data sources and advanced fusion techniques.

To address the limitations imposed by spectral uncertainty, the research community has increasingly focused on fusing HSI with light detection and ranging (LiDAR) data. LiDAR directly captures the 3-D structure of the forest canopy, providing crucial metrics like tree height and crown morphology. This structural information is invaluable for disambiguating species that are spectrally similar but morphologically distinct. Early fusion strategies [9], [10], [11] reflected this principle, though they were largely based on traditional machine learning pipelines. These methods typically involved a multistage process: first, the manual or semiautomated extraction of handcrafted features, such as principal components from spectra and geometric metrics from point clouds, and subsequently training classifiers like support vector machines (SVMs) [9], [10] or random forests [11], [12] on these concatenated feature sets. While pioneering, the efficacy of these approaches was fundamentally limited by their reliance on handcrafted features, which often struggled to capture the complex, nonlinear relationships between the spectral and structural modalities.

The rapid advancement of deep learning has reshaped multisource tree species classification, particularly for integrating HSI and LiDAR data [13]. Unlike traditional approaches, deep learning models can automatically learn hierarchical and representative features directly from raw data, overcoming the limitations of manual feature engineering. Early deep learning efforts demonstrated the superiority of convolutional neural networks (CNNs) over traditional classifiers like SVM and random forests for this task [14]. Subsequent architectural innovations focused on optimizing the fusion of spectral and structural information. These included the development of dual-branch CNN architectures to process each modality separately [15], the introduction of coupled designs to enhance cross-modal interactions [16], and the incorporation of auxiliary subnetworks to extract high-level semantic features for more effective fusion [17]. However, despite these advancements, CNN-based approaches are fundamentally constrained by their local receptive fields. This inherent locality hinders their ability to model the long-range spatial context crucial for characterizing structurally heterogeneous forest canopies, a limitation that has motivated the exploration of architectures with global modeling capabilities [18].

To address this deficiency, Transformer-based architectures were introduced, leveraging self-attention to model global dependencies and facilitate explicit cross-modal feature exchange [19], [20]. However, their inherent emphasis on global semantics often compromises the preservation of fine-grained local details, a critical limitation for precise canopy delineation. This tradeoff has spurred the development of hybrid architectures that pair global models [e.g., Transformer, graph convolutional networks (GCNs)] with CNN to retain local feature fidelity [21], alongside methodologies that explicitly inject high-frequency information to enhance edge representations [22]. Concurrently, the quadratic computational complexity of self-attention has driven the exploration of more efficient backbones, such as state space models (SSMs). The recent HLMamba [23], for instance, leverages the Mamba architecture for efficient long-range dependency modeling and underscores the importance of structural details. Despite these advancements, a critical research gap remains: a unified architecture that can simultaneously achieve global contextual modeling, local detail preservation, and computational efficiency for robust tree species classification in complex forest environments.

The core challenge addressed in this work is fine-grained tree species classification within natural secondary forests, a task severely hindered by complex canopy occlusion and high structural heterogeneity. While prior fusion architectures are promising, their strategies are often generic, resorting to a simple aggregation of local and global features rather than

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a design specifically engineered for this complex scenario. To overcome this limitation, we propose the multisource spectral–spatial Mamba (MS^{3}Mamba) architecture, which introduces a two-level deep fusion strategy that systematically integrates features across both modality (spatial–spectral) and scale (deep–shallow). This hierarchical framework is deliberately engineered for tree species classification. Across modalities, LiDAR provides the crucial structural context of tree height and morphology to disambiguate occluded canopies, thereby refining HSI's spectral signatures. Across scales, shallow features preserve precise crown boundaries for individual tree delineation, while deep features leverage canopy structures and spatial dependencies to distinguish spectrally similar species. The model's learning process is then guided by simultaneous supervision applied at both feature levels, ensuring effective feature fusion in which each component learns a complementary and robust representation. The key contributions of this work are summarized as follows:

1) We propose MS^{3}Mamba, a hierarchical fusion network that deeply integrates multiscale CNN shallow features with Mamba-derived deep spatial–spectral features. This framework establishes a unified spectral–structural representation by leveraging the complementary biochemical and morphological attributes of HSI and LiDAR data, thereby enhancing interclass separability.
2) We design the tree-aware detail refinement (TADR) module to capture fine-grained tree species variations. By combining depth-wise and point-wise convolutions with an affine coupling transform in a cascaded architecture, the module progressively refines canopy boundary details and models local morphological variations, resulting in an improved classification accuracy in challenging natural secondary forest scenes.
3) We establish the MNFPTree-HL dataset, a new, challenging benchmark for fine-grained tree species classification in natural secondary forests. Featuring coregistered, submeter HSI (0.3 m) and LiDAR (0.1 m) data with a comprehensive ground-truth inventory (709 trees, 11 species), it provides a rigorous dataset for validating model robustness under severe canopy occlusion and high species diversity.
4) We conduct comprehensive experiments that validate MS^{3}Mamba's state-of-the-art (SOTA) performance across three distinct datasets. The results confirm our architecture's ability to effectively disentangle spectrally similar species and accurately depict complex canopy boundaries, thereby establishing a robust method for tree species classification in challenging forest environments.

The remainder of this article is organized as follows. Section II reviews the relevant literature. Section III details the study area and our proposed dataset. Section IV provides a comprehensive description of the MS^{3}Mamba architecture. Section V presents and analyzes the experimental results. Finally, Section VI concludes this article and discusses future research directions.

II. Related Works

The relevant literature for this work can be categorized into four primary areas: 1) HSI-based tree species classification; 2) LiDAR-based tree species classification; 3) HSI-LiDAR fusion-based tree species classification; and 4) HSI-LiDAR datasets for tree species classification.

A. HSI-Based Tree Species Classification

HSI has been widely adopted for tree species classification due to its ability to capture fine-grained spectral information from hundreds of narrow, contiguous bands [24]. With the development of deep learning, studies have primarily advanced along three dimensions to maximize the utility of HSI data. First, architectural innovations have sought to improve feature extraction. 3D-CNNs were introduced to jointly learn spectral–spatial features, showing notable improvements over traditional methods [25], [26]. To overcome the limited receptive fields of CNNs, Transformer-based architectures, such as SpectralFormer [27] and hyperspectral Transformers [28], introduced self-attention mechanisms to effectively capture global spectral relationships. Zhang et al. [6] adopted a morphological Transformer to capture spatial characteristics of tree species, while [7] used patch-based and super-resolution methods to better exploit spectral information. Second, to address data scarcity, a significant challenge in forestry, few-shot learning approaches have been proposed. These methods often employ advanced data augmentation strategies, such as proto-maxup, to mitigate overfitting with limited labeled samples [5]. Third, to enhance efficiency, lightweight architectures have been explored, integrating techniques like intelligent band selection to operate on the most informative spectral subsets, thereby reducing data dimensionality without sacrificing accuracy [29]. However, despite these multifaceted advancements, a significant limitation persists across many contemporary HSI-only methods: the tendency to overlook fine-scale local details is crucial for precise boundary delineation. This deficiency is particularly detrimental in dense, heterogeneous forests, where high intraspecies variability and low interspecies similarity reduce model separability and robustness [9].

B. LiDAR-Based Tree Species Classification

As an active remote sensing technology, LiDAR provides high resolution, 3-D structural information, making it invaluable for characterizing canopy height, crown morphology, and branching patterns [30]. Early approaches leverage this data by training traditional machine learning classifiers on handcrafted structural metrics, such as height percentiles and return intensity. While these methods demonstrate potential for distinguishing between broadleaf and coniferous types, their efficacy was constrained by manual feature engineering and coarse classification granularity [31]. To automatically learn more complex geometries, subsequent studies transitioned to end-to-end deep learning models operating directly on point clouds. Architectures like MFFTC-Net enhanced accuracy in species-rich environments by learning multiscale geometric features, yet remained susceptible to data incompleteness caused by occlusion [32]. More advanced methods, such

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as TM2F, pioneered multirepresentation fusion, integrating 3-D point clouds with 2-D projections to leverage robust image-based feature extraction and mitigate data sparsity [33]. Despite these progressively sophisticated geometric methods, they all share a fundamental limitation: the difficulty in distinguishing species that are structurally similar but biochemically distinct. This highlights the absence of spectral information as an intrinsic constraint, thereby motivating the exploration of multimodal strategies that integrate complementary HSI data.

C. HSI-LiDAR Fusion-Based Tree Species Classification

To overcome the inherent limitations of unimodal approaches, the fusion of HSI and LiDAR data has become an active research direction [34]. This synergy is powerful: HSI captures biochemical differences through fine-grained spectral signatures, while LiDAR provides geometric canopy traits, such as height and crown width [35]. The advent of deep learning has catalyzed the development of sophisticated fusion architectures. Early deep learning models primarily utilized dual-branch CNNs to jointly learn spectral–spatial and elevation features, sometimes employing coupled designs with parameter sharing to enhance cross-modal interactions [16]. To overcome the limited receptive fields of pure CNNs, subsequent studies introduced hybrid models. These architectures typically pair a CNN for local feature extraction with a global modeler, such as a Transformer, to capture long-range dependencies, often using cross-attention mechanisms for deep fusion [36]. Further refinements have focused on optimizing the Transformer architecture for multimodal tasks, for instance, by incorporating separable convolutions to improve computational efficiency [20] or by embedding high-frequency details to enhance edge representations [22]. Moving beyond grid-based feature extraction, other hybrid approaches have integrated GCNs to explicitly model global topological relationships derived from LiDAR data [21].

Despite these architectural advancements, existing fusion methods consistently struggle with a fundamental tradeoff. Pure CNN architectures, while efficient, are inherently limited in modeling the global context crucial for distinguishing species based on forest structure. Conversely, architectures designed for Transformer or GCN introduce significant computational complexity, often at the expense of fine-grained local detail preservation. Hybrid approaches have been proposed to mitigate this limitation, yet they often rely on loosely coupled pipelines that remain insufficient for feature fusion. Therefore, a critical research gap remains for a unified, efficient architecture that can simultaneously and synergistically reconcile global contextual modeling with local detail preservation to handle the challenges of complex forest environments.

D. HSI-LiDAR Dataset for Tree Species Classification

A critical limitation of many fusion architectures is their task-agnostic design, which often stems from the datasets used for their development. While several international benchmarks exist, they typically represent ecosystems with substantially different characteristics. For instance, prominent datasets from an Arizona forest [37] and a boreal forest in Finland [14] featured a limited number of species, whereas a dataset from Southern Brazil [38] focused on a tropical environment. The significant variability in species composition and stand structure across these benchmarks fundamentally limits their applicability for models intended for other critical forest types, such as temperate mixed forests. This issue is compounded by the fact that even within our target region of Maoershan, prior local datasets have focused only on a few dominant species, lacking the fine-grained species information required for comprehensive biodiversity assessment [10], [18]. Consequently, the scarcity of public datasets that capture the full species diversity of structurally complex temperate mixed forests represents a significant bottleneck, severely constraining the development and rigorous validation of data-hungry deep learning models. To address this critical research gap, we establish the MNFPTree-HL dataset, a new benchmark specifically designed to drive research in fine-grained classification within these challenging and underrepresented environments.

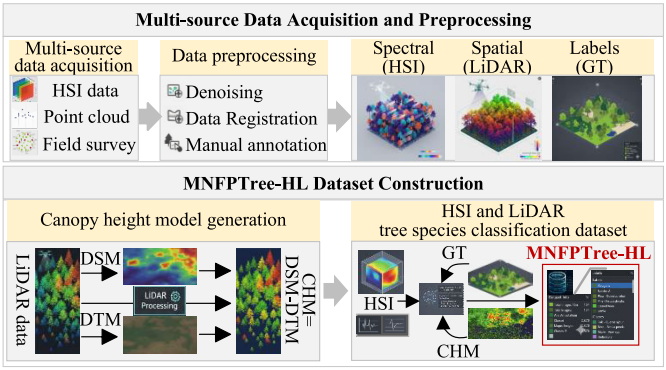


Fig. 2. Flowchart illustrating the construction process of the MNFPTree-HL dataset. The process is divided into two main stages. The first stage, Multisource Data Acquisition and Preprocessing, involves the acquisition and preprocessing of multisource airborne data, including HSI denoising, point cloud denoising, and the co-registration of HSI and LiDAR data to extract spectral, spatial, and annotation information. In the second stage, MNFPTree-HL Dataset Construction, the core dataset is constructed: the raw LiDAR data are processed to generate a CHM, which are then integrated with the preprocessed HSI data and manually annotated ground-truth labels to create the final multisource tree species classification dataset.

III. Study Area and Dataset Construction

This section details the development of the MNFPTree-HL dataset, a custom-collected benchmark created to address the scarcity of public datasets for structurally complex temperate forests. As illustrated in Fig. 2, the construction pipeline involves two primary stages: 1) multisource data acquisition and preprocessing and 2) MNFPTree-HL dataset construction.

A. Study Area

To rigorously assess the model's robustness and practical applicability, we establish our study area within a structurally complex natural secondary forest located in the Maoershan National Forest Park, Heilongjiang Province, China. Situated on the western slope of the Zhangguangcai Range in the Changbai Mountain system, the area lies between

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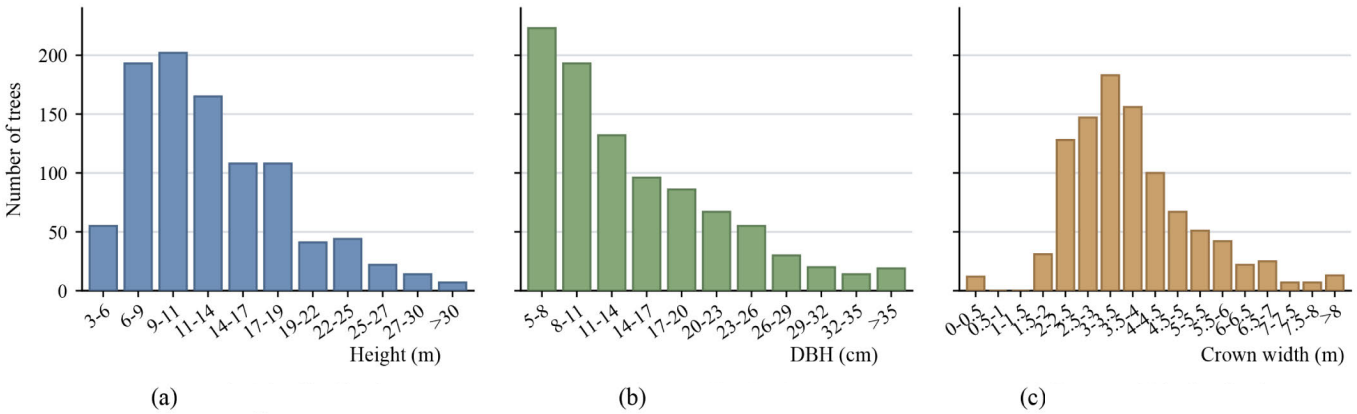


Fig. 3. Forest structure distribution in the MNFPTree-HL dataset. (a) Tree height distribution. (b) Diameter at breast height distribution. (c) Crown width distribution.

45;20–45;25N latitude and 127;30–127;34E longitude. The park is characterized by a mosaic landscape where natural secondary forests are interspersed with plantations, encompassing a total forest area of 23 198.18 hectares with a forest coverage rate of 95.7%.

B. Multisource Data Acquisition and Preprocessing

The MNFPTree-HL dataset comprises three core components: airborne HSI and LiDAR data, alongside a field-surveyed ground-truth inventory.

1) HSI Data Acquisition and Preprocessing: The HSI data is acquired using a large multirotor Uncrewed Aerial Vehicle (UAV), the DJI Fly Car, equipped with an ITRES microSAVI-2 C imaging spectrometer system. This dual-sensor system, consisting of a compact airborne spectrographic imager (CASI) and a shortwave airborne spectrographic imager (SASI), captures data at a 0.3 m spatial resolution across 96 spectral bands with a spectral resolution of 6.2 nm. This high spectral fidelity is vital for resolving the subtle biochemical and biophysical differences among tree species. Before analysis, the raw HSI data undergo a standard denoising procedure to enhance signal quality.

2) LiDAR Data Acquisition and Processing: The LiDAR data are acquired concurrently using a RIEGL VUX-1UAV sensor mounted on the same UAV platform. The resulting point cloud, which provides fine-grained 3-D structural information, is first denoised through filtering. Subsequently, it is processed to generate a digital surface model and a digital terrain model. By calculating the difference between these two models, we derive a canopy height model (CHM) at a 0.1 m spatial resolution. The CHM accurately describes vertical structural attributes, such as tree height and crown morphology, which are indispensable for disentangling the complex and overlapping canopies prevalent in the study area.

3) Multisource Data Co-Registration: To ensure spatial consistency, the hyperspectral and CHM datasets were aligned within the QGIS platform using the field plot survey boundaries. Specifically, both datasets were projected into the unified EPSG:32 652 coordinate system to achieve spatial registration. Concurrently, the CHM data was downsampled to match the spatial resolution of the hyperspectral dataset.

C. MNFPTree-HL Dataset Construction

The ground truth for the MNFPTree-HL dataset is established through a comprehensive, hierarchical field survey. The sampling design encompasses three distinct study sites, within each of which three 40×40 m sample plots were established, resulting in a total of nine plots. Within these plots, we inventory 709 individual trees belonging to 11 distinct species, with their attributes meticulously recorded as detailed in Nomenclature.

To create precise, pixel-level supervision, we then generate the ground-truth map via meticulous manual annotation using the LabelMe tool. This process involves the manual delineation of all 709 individual tree crowns, a procedure rigorously guided by the field survey data. For each tree, the annotation is informed by its unique ID, species, and structural metrics (e.g., H, DBH), with the measured crown extents $(\mathrm{N}, \mathrm{S}, \mathrm{W}, \mathrm{E})$ providing critical geometric constraints for high-fidelity boundary delineation. This detailed, pixel-wise labeling ensures the high-quality supervision essential for training and validating models designed for fine-grained detail preservation. Finally, these ground-truth labels are integrated with the preprocessed HSI data and the LiDAR-derived CHM to form the final multisource dataset.

To provide a more intuitive understanding of the spatial structure of the MNFPTree-HL dataset, we present the distribution histograms of tree height, trunk diameter, and canopy width in Fig. 3, as well as the tree species distribution across the nine sample plots in Fig. 4. The ground survey data indicate that the stand structure is overall stable and demonstrates strong sustainability. The tree height distribution [Fig. 3(a)] exhibits a right-skewed pattern, with medium-height trees (6–14 m) forming the main body, accompanied by a certain proportion of young trees (3–6 m) and a small number of tall individuals (>19 m), reflecting continuous regeneration capacity and the presence of high-value timber resources. The DBH distribution [Fig. 3(b)] follows a typical reverse J-shaped pattern, dominated by small and medium diameter

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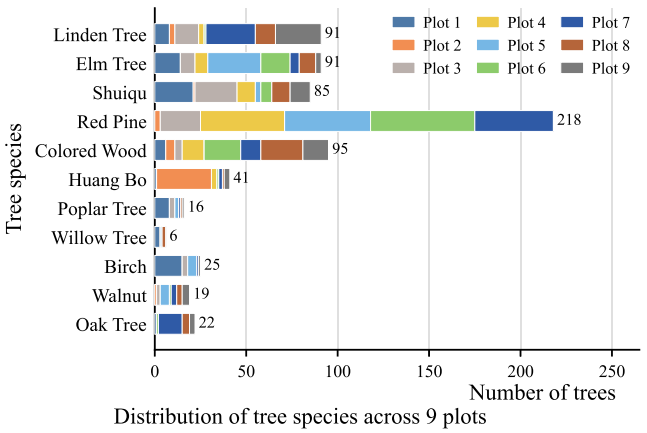


Fig. 4. Forest tree species distribution in the MNFPTree-HL dataset.

classes (5–11 cm), with tree numbers gradually decreasing as DBH increases while showing a slight rebound in the large-diameter class (>35 cm), indicating both strong regeneration ability and a well-maintained multiaged structure. The crown width distribution [Fig. 3(c)] is unimodal and right-skewed, primarily concentrated within the 2.5–4 m range, with a peak at 3–3.5 m, suggesting that medium-sized crowns dominate and that spatial utilization within the stand is efficient.

In terms of species composition, as shown in Fig. 4, among the nine surveyed plots, Red Pine and Colored Wood are the dominant species, with total counts of approximately 220 and 95 trees, respectively, forming the core of the stand. Red Pine is mainly concentrated in plots 5–7, indicating clear habitat preferences, while Colored Wood is abundant in plots 7 and 8. Other species, such as Linden, Elm, and Shuiqu, show moderate abundance and wide distribution across plots. In contrast, companion species, including Oak, Walnut, and Willow, are relatively scarce and occur only sporadically.

Overall, the stand is in a transitional stage from middle-aged to mature forest, characterized by a balanced structure, active regeneration, dominant species prominence, and species diversity, indicating strong ecological stability and significant potential for sustainable management.

IV. Methodology

This section systematically deconstructs the proposed MS^{3}Mamba architecture, as illustrated in Fig. 5. The architecture consists of four sequential components specifically designed for tree species classification: 1) spectral-structural feature encoding (SSFE) that extracts foundational shallow features from HSI and LiDAR data; 2) global canopy context modeling (GCCM) that employs a Mamba-based module to capture long-range morphological and contextual dependencies; 3) TADR that hierarchically fuses the global context with shallow features to enhance fine-grained canopy boundaries; and 4) the classification head (CLH) produces the final pixel-wise tree species classification results. Finally, we conclude with an examination of the hierarchical feature fusion strategy and the composite loss function that guide the model's optimization.

A. SSFE

The inputs to our model are the HSI data, denoted as $\boldsymbol{X} \in \mathbb{R}^{H\times W\times L}$, and the LiDAR-derived CHM, $\boldsymbol{X}_{\text{LiDAR}} \in \mathbb{R}^{H\times W\times 1}$, where H and W are the spatial dimensions and L is the number of HSI bands. To mitigate the high interband correlation and information redundancy inherent in the raw HSI data, we first employ principal component analysis (PCA) for dimensionality reduction. This transformation is designed to retain the most critical discriminative information while suppressing noise, which not only lowers computational complexity, but also enhances model robustness against the curse of dimensionality. The PCA-transformed HSI data, $\boldsymbol{X}_{\text{HSI}}$, can be calculated as follows:

$$\boldsymbol{X}_{\text{HSI}}^{W\times H\times l} = \text{PCA}\left(\boldsymbol{X}^{W\times H\times L}\right) \tag{1}$$

where PCA$(\cdot)$ is the PCA function.

Next, we adopt a patch-based strategy to construct the model's input. For each target pixel, a local neighborhood patch is extracted from both the PCA-transformed HSI data $\boldsymbol{X}_{\text{HSI}}$, capturing the spectral information, and the LiDAR-derived CHM $\boldsymbol{X}_{\text{LiDAR}}$, providing the spatial context. This process yields data patches, $\boldsymbol{X}_{\text{hsi}}$ and $\boldsymbol{X}_{\text{lidar}}$, which encapsulate not only the spectral-structural information of the central pixel but also the spatial context of its surroundings. The input $\boldsymbol{X}_{\text{hsi}}$ and $\boldsymbol{X}_{\text{lidar}}$ are given by

$$\boldsymbol{X}_{\text{hsi}}^{11\times 11\times l} = \text{ExtractPatch}\left(\boldsymbol{X}_{\text{HSI}}^{W\times H\times l}\right) \tag{2}$$

$$\boldsymbol{X}_{\text{lidar}}^{11\times 11\times 1} = \text{ExtractPatch}\left(\boldsymbol{X}_{\text{LiDAR}}^{W\times H\times 1}\right) \tag{3}$$

where $\boldsymbol{X}_{\text{HSI}}^{W\times H\times l}$ and $\boldsymbol{X}_{\text{LiDAR}}^{W\times H\times 1}$ represent the HSI data cube and the single-channel LiDAR data, respectively. W and H are the spatial dimensions, and l denotes the number of principal components retained after applying PCA to the HSI data. The ExtractPatch$(\cdot)$ function extracts the patch of size 11×11 from $\boldsymbol{X}_{\text{HSI}}$ and $\boldsymbol{X}_{\text{LiDAR}}$ for each target pixel and generates patch sets $\boldsymbol{X}_{\text{hsi}}$ and $\boldsymbol{X}_{\text{lidar}}$.

While CNNs are a powerful backbone for extracting local spatial features, they often struggle to effectively integrate complementary information from multimodal sources in a single stream. To address this, our SSFE module is designed to explicitly extract and preserve foundational shallow features from both HSI and LiDAR data, as illustrated in Fig. 6. This is achieved by employing multiscale convolutional operations on each stream to capture hierarchical features, $\boldsymbol{C}_{\text{hsi}}$ and $\boldsymbol{C}_{\text{lidar}}$, and then combining them into a unified shallow feature $\boldsymbol{C}_{\text{shallow}}$. The process is formulated as follows:

$$\boldsymbol{C}_{\text{hsi}} = \text{Cat}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{n\times n}\left(\boldsymbol{X}_{\text{hsi}}\right)\right)\right)\right) \tag{4}$$

$$\boldsymbol{C}_{\text{lidar}} = \text{Cat}\left(\text{ReLU}\left(\text{BN}\left(\text{Conv}_{n\times n}\left(\boldsymbol{X}_{\text{lidar}}\right)\right)\right)\right) \tag{5}$$

$$\boldsymbol{C}_{\text{shallow}} = \text{Conv}\left(\text{Cat}\left(\boldsymbol{C}_{\text{hsi}}, \boldsymbol{C}_{\text{lidar}}\right)\right) \tag{6}$$

where Conv$(\cdot)$ represents the 2-D convolutional operation with a kernel of size n, while BN$(\cdot)$ and ReLU$(\cdot)$ refer to batch normalization and the rectified linear unit activation, respectively. Cat$(\cdot)$ performs a channelwise concatenation. The convolution operation is applied separately to the HSI and LiDAR data using three distinct kernel sizes $n = (3, 5, 7)$, yielding the corresponding outputs $\boldsymbol{C}_{\text{hsi}}$ and $\boldsymbol{C}_{\text{lidar}}$. These features are then

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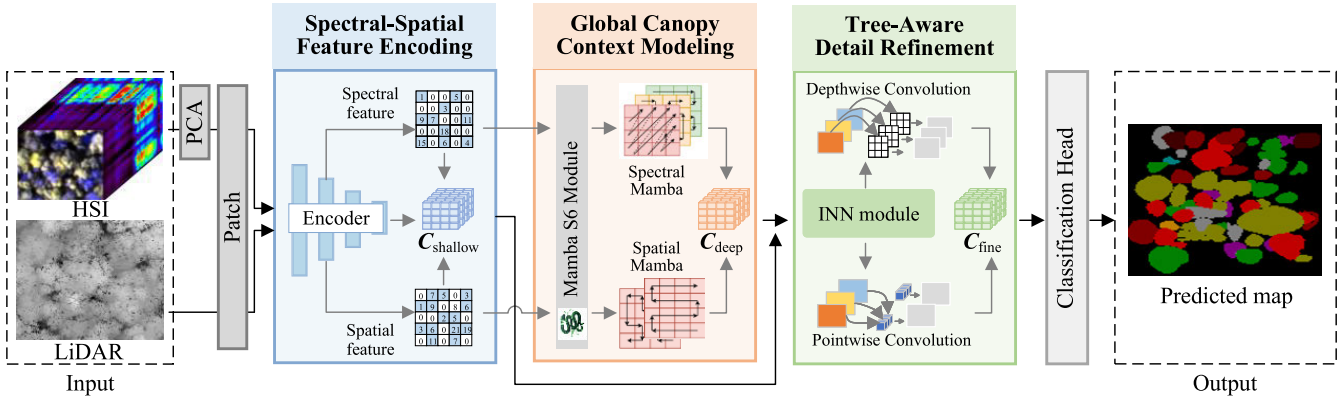


Fig. 5. Overall architecture of MS³Mamba. The model performs PCA on the HSI data, followed by patch extraction. The preprocessed HSI patches and LiDAR patches are then processed through four stages: 1) The SSFE module extracts foundational shallow features. 2) The GCCM module then models long-range dependencies to capture contextual dependencies and get deep features. 3) The TADR module hierarchically fuses these deep contextual features with the shallow details to produce a unified fine representation. 4) The fine representation is fed into a CLH to yield the final pixel-wise classification map.

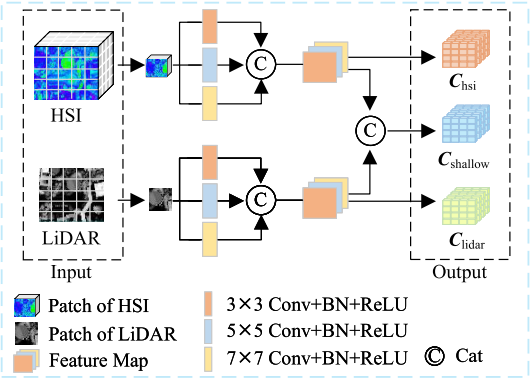


Fig. 6. Architecture of the SSFE module. The HSI and LiDAR inputs are processed in parallel through multiscale convolutional operations with different kernel sizes to extract spectral and spatial features, respectively. These features are then fused to produce the shallow feature.

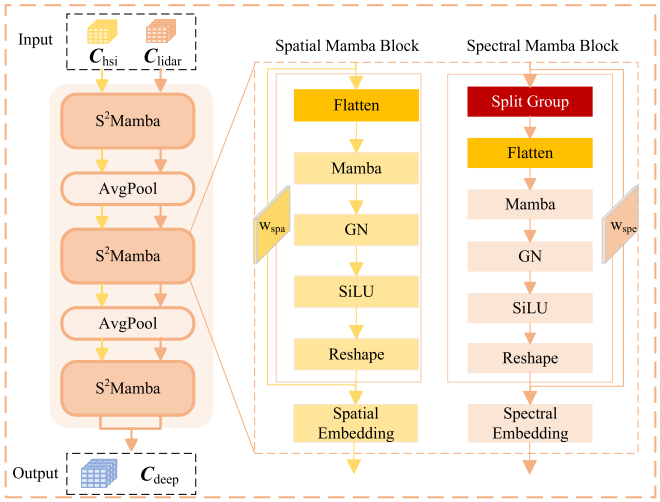


Fig. 7. Architecture of the GCCM. The input data are first processed by parallel spatial and spectral Mamba blocks to obtain spatial and spectral feature embeddings, respectively. The extracted embeddings are then enhanced by integrating the residual information from the input. Finally, a weighted fusion mechanism is employed to aggregate the enhanced Mamba features and residual features, producing the final spatial–spectral fused deep features.

concatenated and passed through a convolutional layer to produce $\boldsymbol{C}_{\text{shallow}}$, which represents the final aggregated shallow features combining both the HSI and LiDAR data.

B. GCCM

After obtaining the modality-specific features $\boldsymbol{C}_{\text{hsi}}$ and $\boldsymbol{C}_{\text{lidar}}$, modeling their global spatial–spectral dependencies becomes essential for characterizing species-specific biochemical and structural attributes across complex canopy environments. While CNNs are inherently constrained by limited local receptive fields and Transformers incur quadratic computational complexity when processing high-resolution data, a more computationally efficient architecture is required to model these long-range dependencies. To this end, we design the GCCM module, which processes the input $\boldsymbol{C}_{\text{hsi}}$ and $\boldsymbol{C}_{\text{lidar}}$ feature streams via separate spatial and spectral Mamba blocks. The detailed architecture of the GCCM is illustrated in Fig. 7.

At its core, Mamba is a sequence-to-sequence model that leverages the principles of SSM. An SSM maps an input sequence $u(t) \in \mathbb{R}^M$ to an output sequence $\boldsymbol{H}^i$ via a latent hidden state $h(t) \in \mathbb{R}^M$. This process is governed by a continuous linear Ordinary Differential Equation (ODE), which is subsequently discretized for computational implementation. This discretization allows the output $\boldsymbol{H}^i$ to be computed efficiently in a convolutional form

$$u_t = \text{Token}(\boldsymbol{C}) \tag{7}$$

$$h_t = \overline{\boldsymbol{A}} h_{t-1} + \overline{\boldsymbol{B}} u_t \tag{8}$$

$$\boldsymbol{H}^i = \overline{\boldsymbol{C}} h_t \tag{9}$$

where the function Token(·) serializes the feature representations from both the HSI and LiDAR branches, i.e., $\boldsymbol{C}$ includes both $\boldsymbol{C}_{\text{hsi}}$ (from the HSI branch) and $\boldsymbol{C}_{\text{lidar}}$ (from the LiDAR branch). These features are separately processed in the HSI and LiDAR branches, capturing the respective spectral and spatial information before being integrated in the following layers. t represents the position of a pixel within the 1-D sequence formed by flattening the input image patch from the HSI and LiDAR data, $u(t)$ is the input feature vector at position t, corresponding to the spatial and spectral information passed

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Algorithm 1 MS^{3}Mamba

Training

Input: HSI data $\boldsymbol{X}^{H\times W\times L}$, LiDAR data $\boldsymbol{X}_{\text{LiDAR}}^{H\times W\times 1}$ and training samples with tree species label $\boldsymbol{P}$

Output: The well-trained model $\mathbf{MS^3Mamba}(\cdot)$

Apply PCA for dimension reduction $\boldsymbol{X}\in\mathbb{R}^{H\times W\times L}$ down to $\boldsymbol{X}_{\text{HSI}}^{H\times W\times l}$, with $l=20$;

Extract patch $\boldsymbol{X}_{\text{HSI}}$ and $\boldsymbol{X}_{\text{LiDAR}}$ with labels determined by the central pixel, generating data patch $\boldsymbol{X}_{\text{hsi}}\in\mathbb{R}^{p\times p\times l}$ and $\boldsymbol{X}_{\text{lidar}}\in\mathbb{R}^{p\times p\times 1}$ with size $p=11$.

for *epoch_num* < *max_epoch_num* **do**

$(\boldsymbol{C}_{\text{shallow}},\boldsymbol{C}_{\text{hsi}},\boldsymbol{C}_{\text{lidar}})\leftarrow \text{SSFE}(\boldsymbol{X}_{\text{hsi}},\boldsymbol{X}_{\text{lidar}})$;

$\boldsymbol{C}_{\text{deep}}\leftarrow \text{GCCM}(\boldsymbol{C}_{\text{hsi}},\boldsymbol{C}_{\text{lidar}})$;

$\boldsymbol{C}_{\text{fine}}\leftarrow \text{TADR}(\boldsymbol{C}_{\text{shallow}},\boldsymbol{C}_{\text{deep}})$;

$\boldsymbol{C}_{\text{cls}}\leftarrow \text{CLH}(\text{Cat}(\boldsymbol{C}_{\text{deep}},\boldsymbol{C}_{\text{fine}}))$;

$\widehat{\boldsymbol{P}}=\text{Argmax}(\text{Softmax}(\text{FC}(\boldsymbol{C}_{\text{cls}})))$;

$\mathcal{L}_{\text{dsf}}\leftarrow\langle \text{MSE}(\boldsymbol{C}_{\text{shallow}},\boldsymbol{C}_{\text{cls}}),\text{MSE}(\boldsymbol{C}_{\text{deep}},\boldsymbol{C}_{\text{fine}})\rangle$;

$\mathcal{L}_{\text{cls}}\leftarrow \text{FocalLoss}(\widehat{\boldsymbol{P}},\boldsymbol{P})$;

$\mathcal{L}=\eta\mathcal{L}_{\text{dsf}}+(1-\eta)\mathcal{L}_{\text{cls}}$;

Backward $\nabla\mathcal{L}$ to update learnable parameters of the $\mathbf{MS^3Mamba}(\cdot)$ model.

end

from the preceding network layer, $h(t)$ denotes the latent hidden state at position t. This state acts as the model's memory, encapsulating a compressed representation of all preceding input information up to that point and $\boldsymbol{H}^i$ is the output feature representation. $\overline{\boldsymbol{A}},\overline{\boldsymbol{B}}$ and $\overline{\boldsymbol{C}}$ are the discretized state matrices that control the evolution of the hidden state $h_{t-1},(h_0=0)$, determines how the current input feature $u(t)$ updates the hidden state $h(t)$, and transform the latent hidden state $h(t)$ into the output feature $\boldsymbol{H}^i$, respectively.

As a pixel-level classification task, tree species identification requires features that capture both spatial context and spectral distinctiveness. To explicitly model these two information streams, we design separate spatial and spectral Mamba modules.

The spatial Mamba module is specifically engineered to model long-range spatial contextual dependencies, producing the output feature $\boldsymbol{H}_{\text{spa}}$ as follows:

$$\boldsymbol{H}_{\text{seq}}=\text{Flatten}\left(\boldsymbol{C}_{\text{lidar}}\right) \tag{10}$$

$$\boldsymbol{H}_{\text{spa}}=\text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\boldsymbol{H}_{\text{seq}}\right)\right)\right)+\boldsymbol{H}^i \tag{11}$$

where $\boldsymbol{C}_{\text{lidar}}\in\mathbb{R}^{B\times p\times p\times D}$ is the spatial feature, B is the batch size, and D is the feature embedding dimension. Each feature in the batch represents a patch of local features. The Flatten$(\cdot)$ operation reshapes the spatial dimensions of the input patch, converting $\boldsymbol{C}_{\text{lidar}}$ into a sequential representation $\boldsymbol{H}_{\text{seq}}\in\mathbb{R}^{B\times p^2\times D}$. This sequence of p^2 now serves as the input to the Mamba layer. The output of the Mamba layer is then passed through a group normalization layer GN$(\cdot)$ and a sigmoid linear unit SiLU$(\cdot)$ activation function to stabilize training and introduce nonlinearity. A residual connection is employed by adding the last input $\boldsymbol{H}^i$ layer. The output of the spatial Mamba module $\boldsymbol{H}_{\text{spa}}\in\mathbb{R}^{B\times p\times p\times D}$ represents the original patch features enriched with long-range spatial contextual information.

The spectral Mamba module is designed to model the fine-grained signatures and relationships across spectral bands, yielding the output feature $\boldsymbol{H}_{\text{spe}}$ as follows:

$$\boldsymbol{H}_{\text{group}}=\text{SplitSpectralGroup}\left(\boldsymbol{C}_{\text{hsi}}\right) \tag{12}$$

$$\boldsymbol{H}_{\text{spe}}=\text{SiLU}\left(\text{GN}\left(\text{Mamba}\left(\text{Flatten}\left(\boldsymbol{H}_{\text{group}}\right)\right)\right)\right)+\boldsymbol{H}^i \tag{13}$$

where $\boldsymbol{C}_{\text{hsi}}$ is the spectral feature, the SplitSpectralGroup$(\cdot)$ function first reshapes the feature dimension D into a 2-D structure of groups and channels-per-group, $\boldsymbol{H}_{\text{group}}\in\mathbb{R}^{B\times p\times p\times G\times M}$ allows the model to focus on relationships within subsets of the spectral bands, G is a hyperparameter representing the number of spectral groups and M is the number of spectral features within each group ($D=G\times M$). $\boldsymbol{H}_{\text{spe}}\in\mathbb{R}^{B\times p\times p\times D}$ is the output feature of the same shape as the input, but now enriched with fine-grained spectral dependency information.

To enhance model robustness, particularly given the limited labeled samples often available for tree species classification, we incorporate residual connections within the GCCM module. Furthermore, the architecture employs an adaptive fusion mechanism to dynamically balance the contributions of the spatial and spectral feature streams. The final deep feature representation, $\boldsymbol{C}_{\text{deep}}$, is thus formulated as

$$\boldsymbol{C}_{\text{deep}}=\boldsymbol{H}^i+\text{w}_{\text{spa}}\cdot\boldsymbol{H}_{\text{spa}}+\text{w}_{\text{spe}}\cdot\boldsymbol{H}_{\text{spe}} \tag{14}$$

where $\boldsymbol{H}^i$ is the original input feature (serving as a global residual connection), while $\boldsymbol{H}_{\text{spa}}$ and $\boldsymbol{H}_{\text{spe}}$ are the outputs of the spatial and spectral Mamba modules, respectively. The learnable scalar weights, w_{spa} and w_{spe} are initialized randomly and updated via backpropagation. These weights allow the model to autonomously determine the relative importance of spatial context versus spectral detail for the tree species classification task. The result $\boldsymbol{C}_{\text{deep}}$ is the final deep feature representation from the GCCM module.

C. TADR

Following the successful extraction of the multiscale shallow features $\boldsymbol{C}_{\text{shallow}}$ and the global contextual features $\boldsymbol{C}_{\text{deep}}$ by the SSFE and GCCM modules, respectively, effective tree species classification requires a synergistic integration of these two complementary feature types. To achieve this, we propose the TADR module, which serves as a global-to-local discriminative framework. Taking $\boldsymbol{C}_{\text{shallow}}$ and $\boldsymbol{C}_{\text{deep}}$ as inputs, the TADR module leverages pointwise and depthwise convolutions to execute affine transformations, thereby enhancing the canopy boundary features of tree species in complex natural secondary forests. The architecture of the TADR is illustrated in Fig. 8.

Inspired by the principle that effective feature fusion must reconcile information across different frequency domains, we employ an inverted residual block [39] as the core mechanism for this integration. Inverted residual block allows for a principled and systematic fusion of low-frequency (shallow/contextual) and high-frequency (deep/detail) features from

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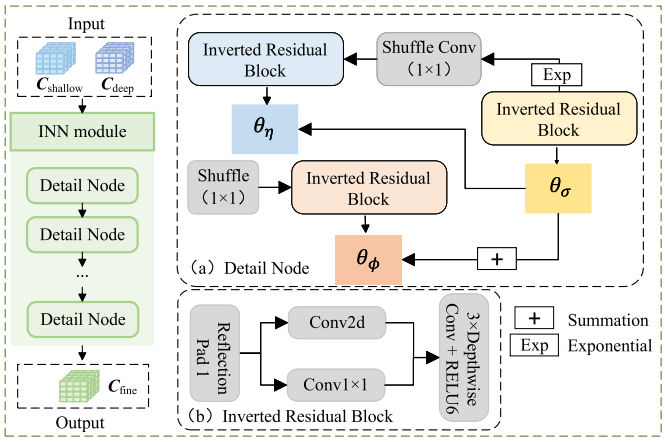


Fig. 8. Architecture of the proposed TADR module. The module is designed as an Invertible Neural Network (INN), consisting of a sequence of Detail Nodes that process shallow features $\boldsymbol{C}_{\text{shallow}}$ and deep features $\boldsymbol{C}_{\text{deep}}$. (a) Each detail node implements an affine coupling layer. It employs three lightweight Inverted Residual Blocks to learn the parameters for a nonlinear mapping θ_ϕ, a translation term θ_σ, and a scaling term θ_η. (b) Inverted Residual Block leverages efficient depthwise and pointwise (1×1) convolutions for feature extraction, analogous to the structure used in MobileNetV2. The stacked detail nodes progressively refine the features, yielding the final fine-grained output$\boldsymbol{C}_{\text{fine}}$.

both HSI and LiDAR data, preserving their respective inductive biases. The resulting fine-grained feature representation, $\boldsymbol{C}_{\text{fine}}$, is formulated as follows:

$$\boldsymbol{Z}_1', \boldsymbol{Z}_2' = \text{Split}\left(\mathcal{F}_{\text{shuffle}}\left(\text{Cat}\left(\boldsymbol{C}_{\text{shallow}}, \boldsymbol{C}_{\text{deep}}\right)\right)\right) \tag{15}$$

$$\boldsymbol{Z}_2 = \boldsymbol{Z}_2' + \theta_\phi\left(\boldsymbol{Z}_1'\right) \tag{16}$$

$$\boldsymbol{Z}_1 = \boldsymbol{Z}_1' \odot \exp\left(\theta_\eta\left(\boldsymbol{Z}_2'\right)\right) + \theta_\sigma\left(\boldsymbol{Z}_2'\right) \tag{17}$$

$$\boldsymbol{C}_{\text{fine}} = \text{Cat}\left(\boldsymbol{Z}_1, \boldsymbol{Z}_2\right) \tag{18}$$

where $\mathcal{F}_{\text{shuffle}}(\cdot)$ represents a learnable channel mixing transformation, implemented as a $\text{Conv}_{1\times1}$, Split$(\cdot)$ divides the channel-mixed feature back into two streams of equal channel dimensions, resulting in the intermediate feature representations $\boldsymbol{Z}_1'$ and $\boldsymbol{Z}_2'$. $\boldsymbol{Z}_2$ is the resulting updated feature stream, which is modulated by the additive coefficient predicted from $\boldsymbol{Z}_1'$, $\boldsymbol{Z}_1$ is the final updated feature stream, which has been scaled and shifted by coefficients derived from $\boldsymbol{Z}_2$. $\theta_\phi(\cdot)$ denotes a nonlinear mapping function, realized by an inverted residual block; $\theta_\sigma(\cdot)$ represents a second distinct inverted residual block that predicts a translation term; $\theta_\eta(\cdot)$ represents a third distinct inverted residual block that predicts a scaling log-factor. The exp$(\cdot)$ is used to ensure that the multiplicative scaling factor is always positive, and the symbol $\odot$ denotes element-wise multiplication. $\boldsymbol{C}_{\text{fine}}$ represents the fine-detail features for the final tree species classification.

D. Classification Head

The final tree species classification feature $\boldsymbol{C}_{\text{cls}}$, is composed of the deep features $\boldsymbol{C}_{\text{deep}}$ and the fine-grained features $\boldsymbol{C}_{\text{fine}}$. From this unified representation, the predicted tree species label $\widehat{\boldsymbol{P}}$ is formulated as follows:

$$\boldsymbol{C}_{\text{cls}} = \text{CLH}\left(\text{Cat}\left(\boldsymbol{C}_{\text{deep}}, \boldsymbol{C}_{\text{fine}}\right)\right) \tag{19}$$

$$\widehat{\boldsymbol{P}} = \text{Argmax}_{j\in[1,c]}\left(\text{Softmax}\left(\text{FC}\left(\boldsymbol{C}_{\text{cls}}\right)\right)\right) \tag{20}$$

where CLH$(\cdot)$ is the CLH, $\boldsymbol{C}_{\text{cls}}$ is the final tree species classification features, Softmax$(\cdot)$ function converts the fusion features into the probability distribution, c represents the number of tree species in the dataset, FC$(\cdot)$ represents fully connected layer, and $\widehat{\boldsymbol{P}}$ corresponds to tree species classification map.

E. Loss Function

To guide the fusion of our multiscale feature streams, we design the deep–shallow feature fusion loss $\mathcal{L}_{\text{dsf}}$, inspired by the hypothesis that low-frequency (shallow) and high-frequency (deep) features should capture complementary information [40]. This loss is engineered with a dual objective: it simultaneously enforces fidelity to the initial shallow details by minimizing their discrepancy with the final fused features while promoting feature disentanglement by maximizing the distance between the intermediate deep and shallow representations. This disciplined optimization guides the model to learn a unified representation that is both detail-rich and composed of complementary features. The deep–shallow feature fusion loss, $\mathcal{L}_{\text{dsf}}$, is as follows:

$$\mathcal{L}_{\text{dsf}} = \frac{\|\boldsymbol{C}_{\text{shallow}} - \boldsymbol{C}_{\text{cls}}\|_2^2}{\left\|\boldsymbol{C}_{\text{deep}} - \boldsymbol{C}_{\text{fine}}\right\|_2^2 + \epsilon} \tag{21}$$

where the numerator $\|\boldsymbol{C}_{\text{shallow}} - \boldsymbol{C}_{\text{cls}}\|_2^2$ is the mean squared error (MSE) between the initial shallow features $\boldsymbol{C}_{\text{shallow}}$ and the final, postfusion features $\boldsymbol{C}_{\text{cls}}$, minimizing this term forces the final representation to retain the critical high-frequency details present in the shallow features. The denominator $\|\boldsymbol{C}_{\text{deep}} - \boldsymbol{C}_{\text{fine}}\|_2^2$ is the MSE between the deep features from the GCCM module $\boldsymbol{C}_{\text{deep}}$ and the fine-detail features from the TADR module $\boldsymbol{C}_{\text{fine}}$. Maximizing this term (by minimizing the loss $\mathcal{L}_{\text{dsf}}$) encourages two streams to learn dissimilar and thus complementary representations. ϵ is a small constant (e.g., 1.01) added for numerical stability to prevent division by zero.

To mitigate the effects of class imbalance, a common challenge in remote sensing where models develop a bias toward easily distinguishable samples, we incorporate the focal loss function [41]. This loss is specifically designed to focus the training process on hard-to-classify examples, thereby improving model generalization. The tree species classification loss, $\mathcal{L}_{\text{cls}}$, is therefore formulated as

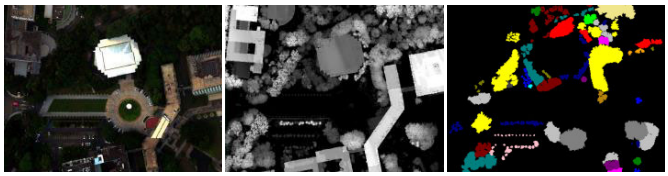
$$\mathcal{L}_{\text{cls}} = -\alpha_{\text{class}}\left(1 - \boldsymbol{p}_{\text{class}}\right)^\gamma log\left(\boldsymbol{p}_{\text{class}}\right) \tag{22}$$

where $\boldsymbol{p}_{\text{class}}$ denotes the predicted probability $\widehat{\boldsymbol{P}}$ of the target class for a pixel, the factor γ is used to focus on challenging samples and is set to two and α_{class} is used to regulate the balance between positive and negative sample loss and is empirically set to one.

The comprehensive loss function, $\mathcal{L}$, is designed to jointly optimize the architecture's dual objectives. It is a composite of the deep–shallow feature fusion loss $\mathcal{L}_{\text{dsf}}$, which enforces a well-structured feature hierarchy, and the tree species classification loss $\mathcal{L}_{\text{cls}}$, which guides the model toward a discriminative final prediction. This composite loss $\mathcal{L}$ is given by

$$\mathcal{L} = \eta\mathcal{L}_{\text{dsf}} + (1 - \eta)\,\mathcal{L}_{\text{cls}} \tag{23}$$

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(a) (b) (c)

Number	Tree Species	Trees	Pixel	Color
1	Terminalia arjuna	9	2802385	
2	Kigelia africana	1	517046	
3	Roystonea regia	25	578928	
4	Ficus macrophylla	33	10613149	
5	Delonix regia	2	597083	
6	Archontophoenix alexandrae	1	70303	
7	Litchi chinensis	31	3616393	
8	Mangifera indica	4	587487	
9	Araucaria cunninghamii	36	2285249	
10	Livistona chinensis	6	349851	
11	Dracontomelon	7	803545	
12	Acacia auriculiformis	6	2230821	
13	Ficus concinna	15	5017957	
14	Ficus elastica	2	4269403	
15	Leucaena leucocephala	1	137162	
16	Bauhinia purpurea	3	223554	
17	Camphora officinarm	27	964668	
18	Dimocarpus longan	15	2079575	
19	Bombax ceiba	3	389571	
20	Corymbia citriodora	8	2241563	

(d)

Fig. 9. SZUTreeData-R1 dataset. (a) HSI false color image. (b) CHM map. (c) Ground truth. (d) Category and label samples.



(a) (b) (c)

Number	Tree Species	Trees	Pixel	Color
1	Terminalia arjuna	16	4676843	
2	Kigelia africana	2	145603	
3	Roystonea regia	2	351353	
4	Ficus macrophylla	18	5566268	
5	Delonix regia	16	4919621	
6	Archontophoenix alexandrae	2	52749	
7	Litchi chinensis	60	5843349	
8	Mangifera indica	2	223778	
9	Araucaria cunninghamii	6	333493	
10	Livistona chinensis	12	520990	
11	Dracontomelon	8	1244521	
12	Acacia auriculiformis	5	1043262	
13	Ficus concinna	32	10148927	
14	Ficus elastica	1	1013809	
15	Leucaena leucocephala	1	106293	
16	Bauhinia purpurea	27	1808259	
17	Melaleuca	31	3803135	
18	Ficus virens	30	7227444	
19	Plumeria rubra	5	332495	
20	Casualina equisetifolia	5	878456	
21	Swietenia mahagoni	15	7403467	

(d)

Fig. 10. SZUTreeData-R2 dataset. (a) HSI false color image. (b) CHM map. (c) Ground truth. (d) Category and label samples.

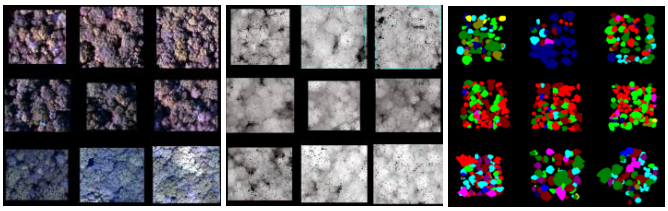
where η is a hyperparameter that balances the contribution of the two loss objectives. Based on empirical validation, we set $\eta = 0.8$. In summary, the composite loss function $\mathcal{L}$ is designed to create a synergistic training objective. It simultaneously enforces robust, disentangled feature learning via $\mathcal{L}_{\text{dsf}}$ while directly optimizing for tree species classification accuracy through $\mathcal{L}_{\text{cls}}$. This dual-objective optimization guides the end-to-end architecture to learn a unified representation that simultaneously captures global species context and preserves fine-grained local boundaries, enabling precise tree species classification.

V. Experiments

A. Data Description

Our experiments are conducted on three distinct datasets: two established public datasets, SZUTreeData-R1 [2], [4] and SZUTreeData-R2 [42], [43], for comparative analysis against existing literature, and our custom-collected MNFPTree-HL dataset to rigorously evaluate model robustness in a challenging real-world scenario. Visualizations of the raw data, ground-truth maps, and data partitions for all three datasets are provided in Figs. 9–11.

1) SZUTreeData-R1 Dataset: The SZUTreeData-R1 dataset, collected over the Yu Hai Campus of Shenzhen University, has a spatial dimension of 3085 × 2.405 pixels. It encompasses 20 tree species and 235 individual trees. These species are: *Terminalia arjuna, Kigelia africana, Roystonea regia, Ficus macrophylla, Delonix regia, Archontophoenix alexandrae, Litchi chinensis, Mangifera indica, Araucaria cunninghamii, Livistona chinensis, Dracontomelon, Acacia*



(a) (b) (c)

Number	Tree Species	Trees	Pixel	Color
1	Linden Tree	91	11789	
2	Elm Tree	91	12267	
3	Shuiqu	85	1408	
4	Red Pine	218	14947	
5	Colored Wood	95	13837	
6	Huang Bo	41	8129	
7	Poplar Tree	16	2713	
8	Willow Tree	6	617	
9	Birch	25	8089	
10	Walnut	19	2430	
11	Oak Tree	22	3463	

(d)

Fig. 11. MNFPTree-HL dataset. (a) HSI false color image. (b) CHM map. (c) Ground truth. (d) Category and label samples.

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TABLE I
QUANTITATIVE PER-CLASS ACCURACY RESULTS OF SZUTREEDATA-R1 DATASET. BEST IS BOLD AND SECOND WITH UNDERLINE

Class	CoupledCNN [16]	MAHiDFNet [44]	HCTnet [36]	CALC [17]	MFT [19]	MVit [20]	NCGFL2 [40]	IDnet [22]	S3F2Net [21]	HLMamba [23]	MS^{3}Mamba (proposed)
Terminalia arjuna	65.25	95.99	97.21	85.72	96.12	95.77	97.21	96.57	93.51	96.50	**97.26**
Kigelia africana	14.47	64.32	89.91	51.58	87.99	81.41	89.66	87.36	74.15	86.31	**89.92**
Roystonea regia	57.32	83.14	89.64	84.07	89.18	86.39	88.96	89.42	79.60	87.20	**91.02**
Ficus macrophylla	78.40	98.59	98.12	90.04	97.20	97.72	98.56	98.62	93.89	98.87	**98.81**
Delonix regia	75.66	78.87	93.55	65.81	88.51	85.04	94.02	88.13	81.15	92.96	**94.36**
Archontophoenix alexandrae	38.10	67.53	70.79	55.31	66.17	58.60	**71.29**	70.75	57.60	65.46	69.15
Litchi chinensis	92.65	94.96	98.02	98.13	96.81	94.90	98.01	96.94	96.17	98.07	**98.72**
Mangifera indica	44.03	86.51	95.06	68.55	94.7	90.36	94.95	94.62	85.46	94.48	**95.84**
Araucaria cunninghamii	79.96	96.47	98.36	95.90	97.71	96.38	97.77	97.61	95.04	97.49	**98.57**
Livistona chinensis	55.70	79.49	93.72	63.86	91.36	87.29	93.13	93.23	69.13	91.24	**94.76**
Dracontomelon	35.03	50.27	95.89	71.18	93.51	92.33	96.58	90.21	77.95	91.40	**97.45**
Acacia auriculiformis	80.23	97.62	**98.86**	91.01	98.17	96.85	98.74	98.46	94.37	97.19	98.66
Ficus concinna	94.55	97.27	99.13	85.30	96.95	94.10	98.44	98.16	96.28	97.56	**99.14**
Ficus elastica	79.27	89.97	99.05	97.27	98.87	98.00	99.03	98.73	97.09	97.92	**99.40**
Leucaena leucocephala	39.30	83.31	90.61	60.21	91.59	85.04	**93.91**	90.71	79.92	87.01	92.27
Bauhinia purpurea	61.68	73.71	**96.91**	59.35	94.60	91.03	95.45	95.90	89.51	92.99	96.81
Camphora officinarum	78.35	58.83	98.42	90.43	97.64	91.63	96.74	97.56	88.28	93.77	**97.82**
Dimocarpus longan	62.18	95.33	98.37	57.86	96.44	93.34	98.72	98.10	88.91	96.78	**98.75**
Bombax ceiba	53.12	87.95	99.05	91.41	98.98	97.83	99.21	99.11	96.70	98.88	**99.22**
Corymbia citriodora	40.26	60.38	99.33	82.15	99.49	97.82	99.45	99.56	98.26	99.38	**99.80**
OA(%)	74.61	73.03	98.14	86.34	96.78	95.35	97.86	97.39	93.17	97.12	**98.36**
AA(%)	59.78	48.92	95.07	71.48	93.60	90.59	94.99	94.00	86.65	93.07	**95.47**
Kappa(%)	70.92	68.31	97.88	84.43	96.34	94.70	97.56	97.02	92.21	96.72	**98.13**

TABLE II
QUANTITATIVE PER-CLASS ACCURACY RESULTS OF SZUTREEDATA-R2 DATASET. BEST IS BOLD AND SECOND WITH UNDERLINE

Class	CoupledCNN [16]	MAHiDFNet [44]	HCTnet [36]	CALC [17]	MFT [19]	MVit [20]	NCGFL2 [40]	IDnet [22]	S3F2Net [21]	HLMamba [23]	MS^{3}Mamba (proposed)
Terminalia arjuna	87.09	92.18	96.13	82.18	94.29	91.40	**96.38**	93.20	88.36	95.21	96.20
Kigelia africana	34.45	57.82	59.75	63.17	67.28	54.98	66.32	60.72	50.15	56.97	**67.53**
Roystonea regia	41.52	67.93	80.19	49.94	81.65	72.24	80.43	78.78	68.23	82.09	**81.74**
Ficus macrophylla	68.62	90.01	**97.31**	89.09	93.63	87.20	95.02	95.78	83.43	96.28	96.60
Delonix regia	91.33	97.03	97.43	92.83	97.11	94.30	97.21	95.73	94.04	94.67	**98.15**
Archontophoenix alexandrae	**54.53**	30.27	47.98	43.97	44.71	22.84	47.46	37.10	39.13	49.28	46.16
Litchi chinensis	74.94	96.13	97.99	95.83	97.62	95.81	97.20	**98.27**	96.49	97.63	97.98
Mangifera indica	30.26	74.69	86.13	60.73	84.13	72.87	83.49	**86.57**	65.72	85.94	85.36
Araucaria cunninghamii	33.66	79.82	88.03	52.19	83.17	74.01	85.11	81.57	67.41	87.41	**88.87**
Livistona chinensis	57.02	78.80	**91.54**	80.94	90.75	85.54	91.44	90.92	80.36	89.83	91.09
Dracontomelon	67.92	78.43	96.29	85.77	**96.53**	90.28	96.16	94.51	84.80	93.45	94.26
Acacia auriculiformis	26.05	81.95	94.75	68.35	89.91	82.84	92.29	93.13	68.61	94.30	**94.99**
Ficus concinna	84.96	93.01	97.00	93.38	95.86	91.70	**98.12**	95.76	86.44	96.92	97.39
Ficus elastica	71.47	89.51	96.55	89.32	95.95	91.03	96.33	95.68	84.99	95.79	**96.99**
Leucaena leucocephala	50.36	67.37	74.69	71.27	70.79	65.16	**81.12**	76.83	34.86	69.92	80.45
Bauhinia	57.20	63.57	96.19	88.93	93.20	87.45	95.72	93.89	81.23	94.59	**97.86**
Melaleuca	87.97	97.66	99.06	94.53	98.85	97.49	98.57	**99.19**	97.50	98.81	98.90
Ficus virens	65.54	90.68	97.10	89.04	96.57	94.16	97.09	97.21	90.57	96.34	**98.65**
Plumeria rubra	81.91	92.39	96.99	90.01	96.78	95.96	**97.41**	96.52	94.32	95.47	97.24
Casuarina equisetifolia	35.63	88.71	**98.76**	81.13	97.36	95.07	96.78	97.04	93.81	97.28	96.80
Swietenia mahagoni	73.07	95.79	98.25	87.17	96.46	93.61	97.64	96.64	91.54	96.44	**98.89**
OA(%)	74.21	81.04	97.02	88.20	95.50	91.87	96.51	95.66	88.71	95.82	**97.11**
AA(%)	52.90	39.59	90.01	67.17	88.69	82.66	89.87	88.34	76.76	88.79	**90.35**
Kappa(%)	70.96	78.59	96.39	86.81	94.99	90.93	96.11	95.16	87.40	95.34	**96.67**

auriculiformis, Ficus concinna, Ficus elastica, Leucaena leucocephala, Bauhinia purpurea, Camphora officinarum, Dimocarpus longan, Bombax ceiba, and Corymbia citriodora. The dataset features co-registered HSI and LiDAR data, both at a 0.1 m spatial resolution. The HSI data, acquired by a UAV, consists of 98 noise-reduced bands (399–924 nm) with a 5.5 nm spectral resolution, while the CHM is derived from backpack LiDAR. Visualizations of the raw imagery, ground-truth map, and data partitions are provided in Fig. 9.

2) SZUTreeData-R2 Dataset: The SZUTreeData-R2 dataset serves as an expanded version of R1, collected over the same area and sharing identical sensor specifications (0.1 m HSI and CHM resolution, 98 bands, 399–924 nm). It differs in its larger spatial extent of 4040 $\times$ 2.444 pixels and a greater sample size, encompassing 296 individual trees across 21 species. These species are: *Terminalia arjuna, Kigelia africana, Roystonea regia, Ficus macrophylla, Delonix regia, Archontophoenix alexandrae, Litchi chinensis, Mangifera indica, Araucaria cunninghamii, Livistona chinensis, Dracontomelon, Acacia auriculiformis, Ficus concinna, Ficus elastica, Leucaena leucocephala, Bauhinia purpurea, Melaleuca, Ficus virens, Plumeria rubra, Casuarina*

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TABLE III
QUANTITATIVE PER-CLASS ACCURACY RESULTS ON THE MNFPTREE-HL DATASET. BEST IS BOLD AND SECOND WITH UNDERLINE

Class	CoupledCNN [16]	MAHiDFNet [44]	HCTnet [36]	CALC [17]	MFT [19]	MVit [20]	NCGFL2 [40]	IDnet [22]	S3F2Net [21]	HLMamba [23]	MS^{3}Mamba (proposed)
Linden Tree	68.49	80.25	**90.71**	59.49	63.38	61.77	89.39	65.05	69.88	62.29	86.86
Elm Tree	34.23	72.82	81.67	62.75	59.05	47.06	82.65	64.93	52.31	45.36	**90.90**
Shuiqu	7.05	0.00	83.21	0.00	67.72	27.18	82.65	49.72	21.89	39.89	**91.56**
Red Pine	0.00	0.00	83.30	0.00	35.73	23.04	**83.51**	28.75	0.42	31.71	68.08
Colored Wood	25.07	0.00	84.71	42.81	76.26	54.17	85.39	71.54	51.04	50.74	**87.69**
Huang Bo	31.85	47.05	84.02	53.22	59.95	47.53	83.70	65.77	38.29	49.53	**88.60**
Poplar Tree	37.37	73.76	86.39	50.88	59.57	58.76	84.48	71.28	54.24	53.44	**91.68**
Willow Tree	55.11	78.44	89.42	66.67	78.99	71.97	89.96	75.24	77.64	74.13	**91.83**
Birch	81.58	87.12	92.31	74.08	85.51	79.04	94.47	86.31	85.98	77.28	**95.44**
Walnut	38.14	0.00	85.15	46.26	65.25	50.51	82.68	53.95	56.54	61.70	**87.20**
Oak Tree	0.63	0.00	82.37	38.34	66.62	17.00	79.34	35.36	22.45	32.80	**86.99**
OA(%)	47.16	64.44	87.06	57.37	67.35	58.09	86.78	68.37	60.40	58.54	**90.10**
AA(%)	34.50	39.95	85.75	44.96	65.27	48.91	85.29	60.72	48.24	52.62	**87.89**
Kappa(%)	38.03	58.04	85.02	50.48	62.21	51.14	84.70	63.26	53.69	51.74	**88.55**

equisetifolia, and Swietenia mahagoni. Further details and visualizations of the raw imagery, ground-truth map, and data partitions are provided in Fig. 10.

3) MNFPTree-HL Dataset: Compared with benchmark datasets typically sourced from managed environments like university campuses, the natural forest of Maoershan presents a far more severe challenge due to its high species diversity, complex stand structure, and severe canopy occlusion. This diversity is exemplified by 11 co-existing tree species: Linden Tree, Elm Tree, Shuiqu, Red Pine, Colored Wood, Huang Bo, Poplar Tree, Willow Tree, Birch, Walnut, and Oak Tree. These tree species characteristics introduce substantial spatial ambiguity and intraspecies variability, creating a more realistic and demanding classification scenario. Therefore, a successful performance on this dataset is not merely an incremental improvement but a critical validation of a model's robustness and operational readiness for real-world applications. The inherent complexity of this dataset thus provides a rigorous testbed for evaluating the effectiveness of our proposed fusion architecture. Visualizations of the raw imagery, ground-truth map, and class-wise pixel statistics for the dataset are provided in Fig. 11.

B. Experimental Settings

1) Experimental Details: All experiments are implemented using the PyTorch framework and conducted on a server equipped with an Intel Xeon Gold 6330 CPU and an NVIDIA RTX 4090 GPU with 24 GB of VRAM. For the proposed MS^{3}Mamba architecture, we employ the Adam optimizer with an initial learning rate of 1×10^{-4}. The model is trained for 80 epochs with a batch size of 256.

The dataset is split into 10% training and 90% testing samples without overlap. Here, "no overlap" indicates that the central pixels used for training and testing are mutually exclusive, while the constructed patches may share neighboring spatial context. This patch-based strategy is widely adopted in hyperspectral tree species classification to preserve local spatial continuity and canopy structure. Due to the limited dataset size and uneven spatial distribution of tree species, spatially disjoint partitioning (i.e., using geographically separated regions for training and testing) is not adopted in this study. Nevertheless, all compared methods follow the same data splitting protocol to ensure fair evaluation.

2) Evaluation Metrics: Model performance is evaluated using standard quantitative metrics: overall accuracy (OA), average accuracy (AA), and Cohen's Kappa coefficient, supplemented by per-class accuracy. In all result tables, the best-performing results are highlighted in bold, while the second-best results are underlined. For qualitative analysis, we also generate the corresponding classification maps.

3) Comparative Methods: To comprehensively evaluate the performance of MS^{3}Mamba, we conducted a comparative analysis against ten SOTA multimodal fusion methods. The selected methods represent a diverse range of architectural paradigms, including those based on CNN, Transformer, and hybrid designs: CoupledCNN [16], MAHiDFNet [44], HCTNet [36], CALC [17], MFT [19], MViT [20], NCGFL2 [40], IDNet [22], S3F2Net [21], and HLMamba [23]. Model performance is evaluated through both qualitative visualization and quantitative metrics.

C. Comparison With SOTA Methods

This section presents a comparative analysis of the proposed MS^{3}Mamba architecture against the selected SOTA methods on all three datasets. The quantitative evaluation, comprising OA, AA, the Kappa coefficient, and per-class accuracies, is summarized in Tables I–III. For qualitative assessment, the corresponding tree species classification maps are visually presented in Figs. 12–14.

1) SZUTreeData-R1 Dataset: As presented in Table I, MS^{3}Mamba achieves the highest overall metrics (OA, AA, and Kappa) on the SZUTreeData-R1 dataset. This quantitative superiority is further reflected at the class level, where our model outperforms all competitors on 16 of the 20 species. For the remaining few classes, such as *Leucaena leucocephala* and *Bauhinia purpurea*, MS^{3}Mamba remains highly competitive, securing a close second place with only marginal performance gaps. Notably, even for challenging classes, such as *Archontophoenix alexandrae*, where all methods struggle due to complex tree structures, our model remains competitive, trailing the top performer by only two percentage points.

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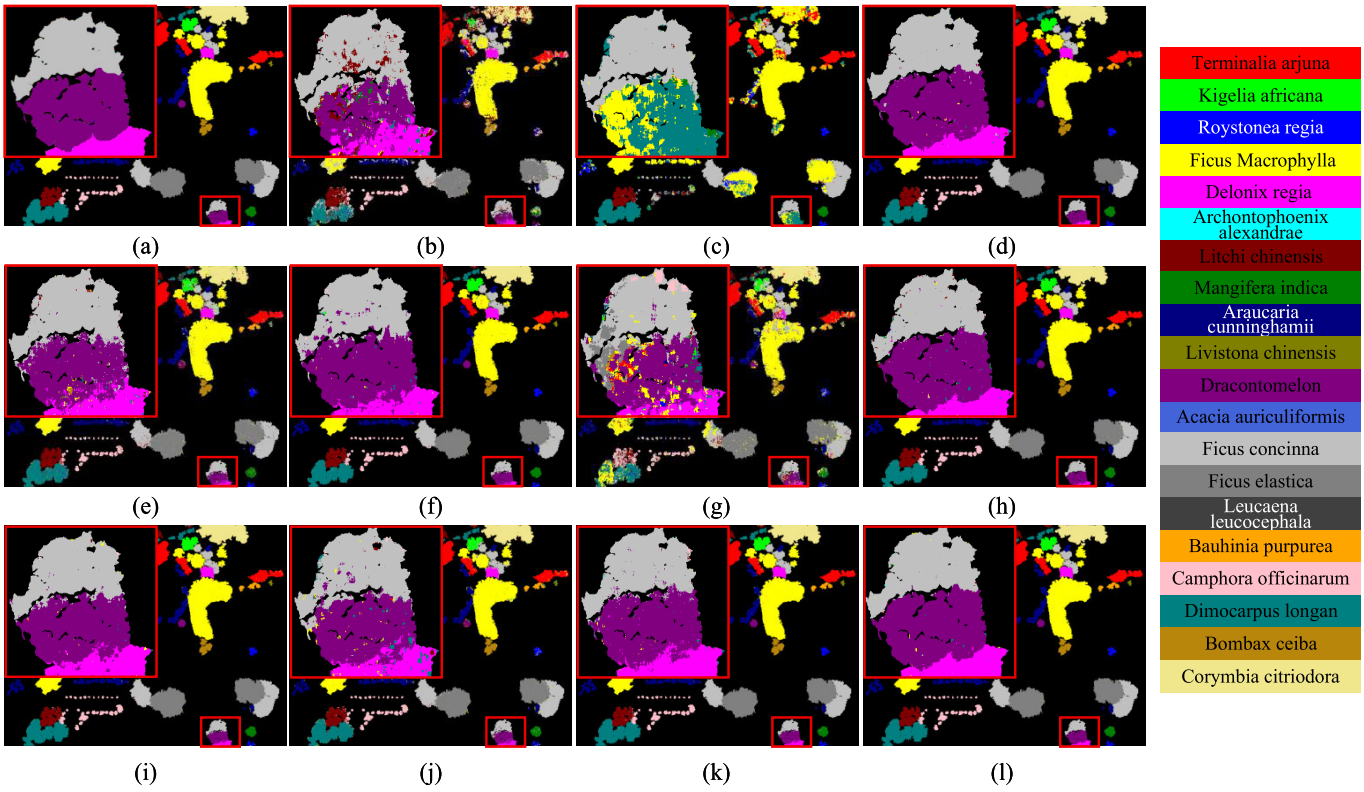


Fig. 12. Qualitative visualization of the tree species classification map for SZUTreeData-R1 dataset. (a) GT. (b) CoupledCNN [16]. (c) MAHiDFNet [44]. (d) HCTnet [36]. (e) CALC [17]. (f) MFT [19]. (g) MVit [20]. (h) NCGFL2 [40]. (i) IDnet [22]. (j) S3F2Net [21]. (k) HLMamba [23]. (l) MS^{3}Mamba.

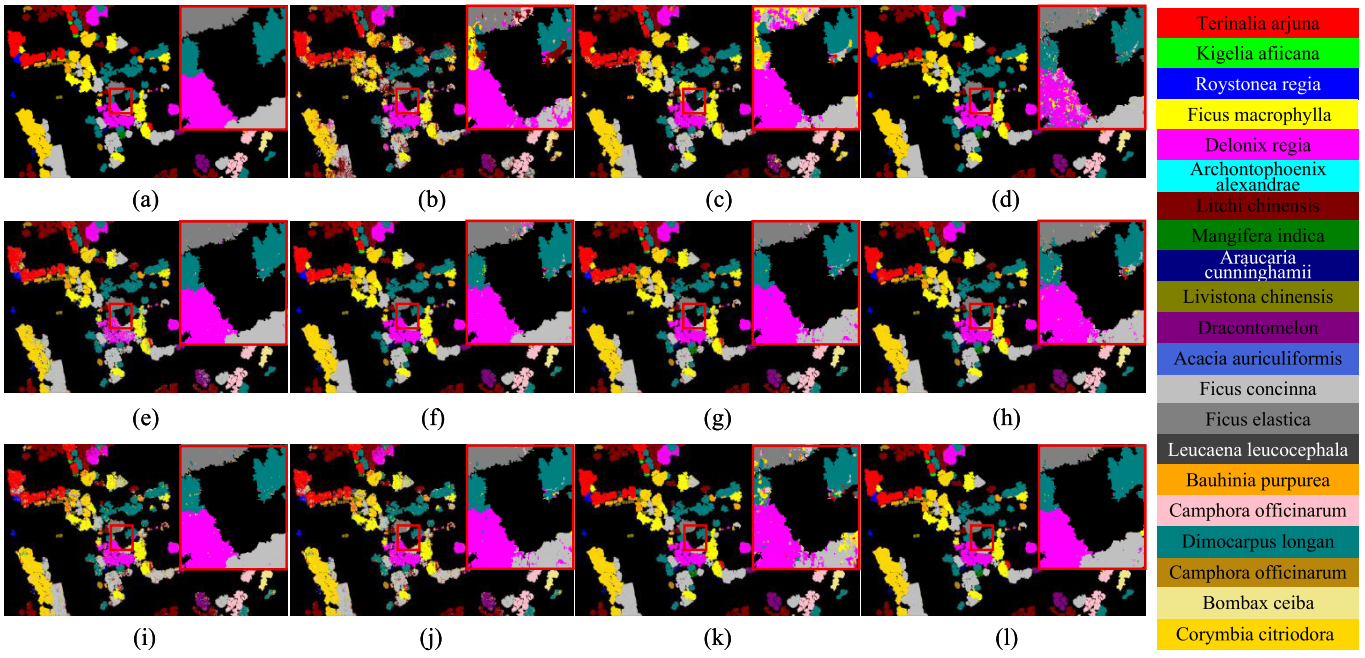


Fig. 13. Qualitative visualization of the tree species classification map for the SZUTreeData-R2 dataset. (a) GT. (b) CoupledCNN [16]. (c) MAHiDFNet [44]. (d) HCTnet [36]. (e) CALC [17]. (f) MFT [19]. (g) MVit [20]. (h) NCGFL2 [40]. (i) IDnet [22]. (j) S3F2Net [21]. (k) HLMamba [23]. (l) MS^{3}Mamba.

The qualitative results, visualized in Fig. 12, corroborate the quantitative findings. Many comparative methods, such as CoupledCNN and S3F2Net, produce classification maps with significant "salt-and-pepper" noise, particularly in homogeneous regions. Others, like MAHiDFNet, exhibit large-scale, spatially contiguous misclassifications. In stark contrast, the

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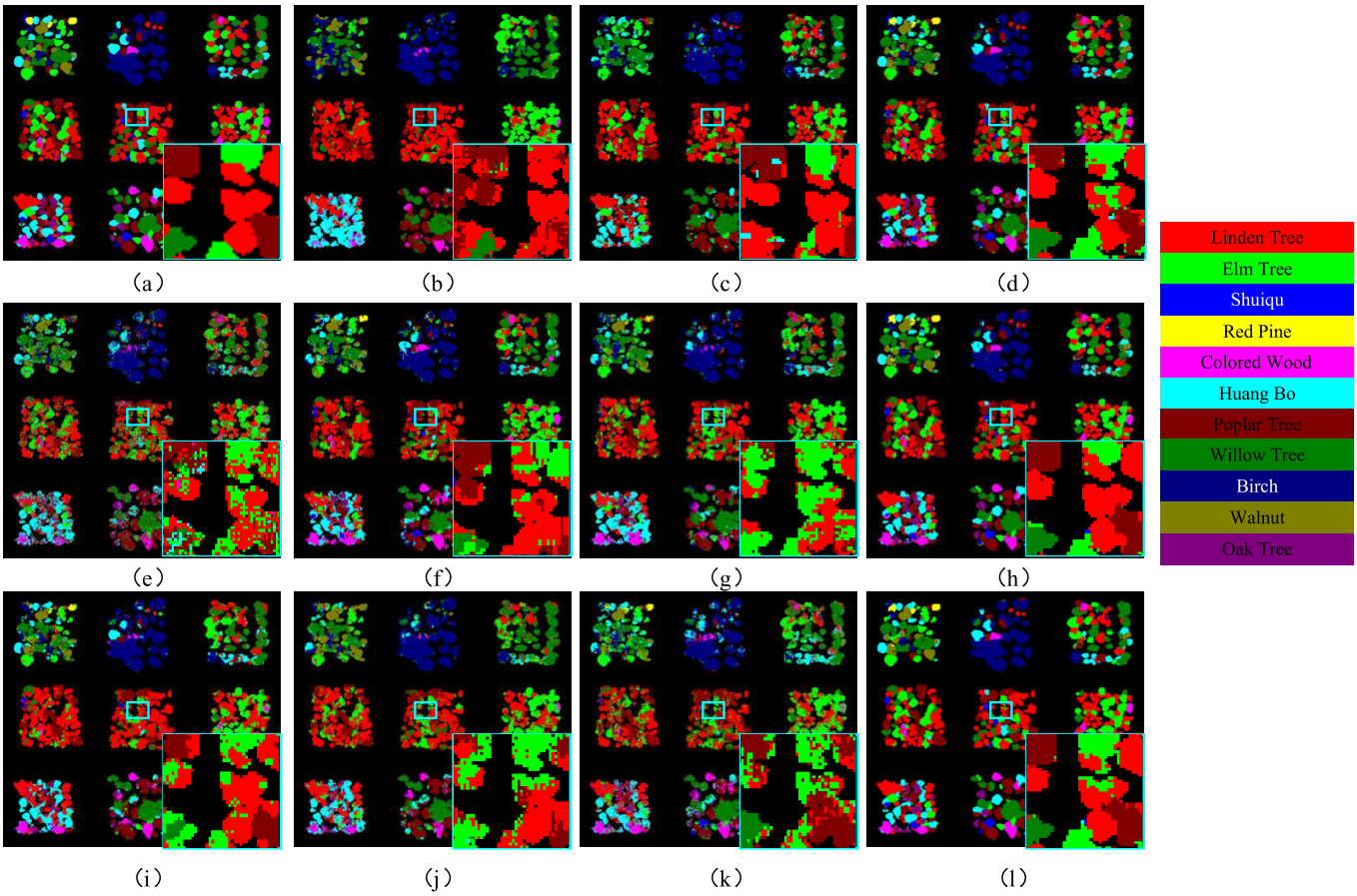


Fig. 14. Qualitative visualization of the tree species classification map for the MNFPTree-HL dataset. (a) GT. (b) CoupledCNN [16]. (c) MAHiDFNet [44]. (d) HCTnet [36]. (e) CALC [17]. (f) MFT [19]. (g) MVit [20]. (h) NCGFL2 [40]. (i) IDnet [22]. (j) S3F2Net [21]. (k) HLMamba [23]. (l) MS^{3}Mamba.

map generated by MS^{3}Mamba exhibits significantly higher visual fidelity to the ground truth, producing spatially coherent regions and well-defined canopy boundaries. This visual superiority directly reflects the model's effectiveness in reconciling the species global context with tree local detail, validating the architectural design.

2) SZUTreeData-R2 Dataset: The comparative results on the SZUTreeData-R2 dataset are presented in Table II. MS^{3}Mamba again achieves the highest overall metrics (OA, AA, and Kappa), demonstrating its robust performance on a larger and more diverse sample set. At the class level, our model secures either the top or second-best performance in 14 out of 21 species. Even in highly competitive categories like *Litchi chinensis* and *Mangifera indica*, MS^{3}Mamba trails the leading method by a marginal gap of less than 1%. While all methods struggle with the structurally complex *Archontophoenix alexandrae* class, this consistent high performance across the majority of species underscores the model's strong generalization capabilities.

The qualitative results in Fig. 13 further reinforce these findings. The classification maps of several comparative methods, including MViT and HLMamba, are marred by "salt-and-pepper" noise and fragmentation within homogeneous regions. MAHiDFNet, in particular, exhibits large-scale spatial misclassifications, failing to preserve structural integrity. In stark contrast, MS^{3}Mamba produces a map with superior spatial coherence and cleaner class boundaries, which is visually the most faithful to the ground truth. This visual evidence highlights the model's effectiveness in leveraging global context to ensure local consistency, a key advantage of our hierarchical fusion design.

3) MNFPTree-HL Dataset: The superiority of MS^{3}Mamba is most evident on our challenging, custom-collected MNFPTree-HL dataset, as shown in Table III. Our model achieves the highest overall metrics (OA, AA, and Kappa) and secures the top accuracy in 9 out of 11 species. In stark contrast, a significant number of SOTA methods fail catastrophically in this complex natural forest environment, with classification accuracies plummeting below 50% for multiple species and even dropping to zero in some cases. While MS^{3}Mamba also faces challenges with the most difficult classes like Linden Tree and Red Pine, its ability to maintain highly competitive performance across the board underscores its robustness and superior generalization in the real world.

The qualitative results in Fig. 14 visually affirm these quantitative findings. The classification maps of many SOTA methods, including MViT and HLMamba, are severely degraded by the "salt-and-pepper" effect and fail to preserve the integrity of homogeneous tree stands. In contrast, MS^{3}Mamba produces a map that is spatially coherent and

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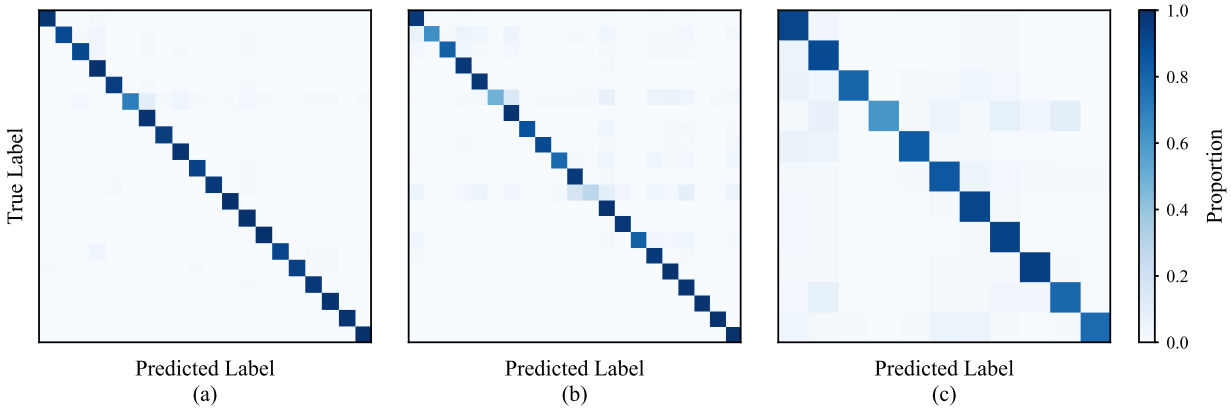


Fig. 15. Confusion matrices of the classification results on three datasets: (a) SZUTreeData-R1, (b) SZUTreeData-R2, and (c) MNFPTree-HL dataset. The species names are as those in Figs. 9–11.

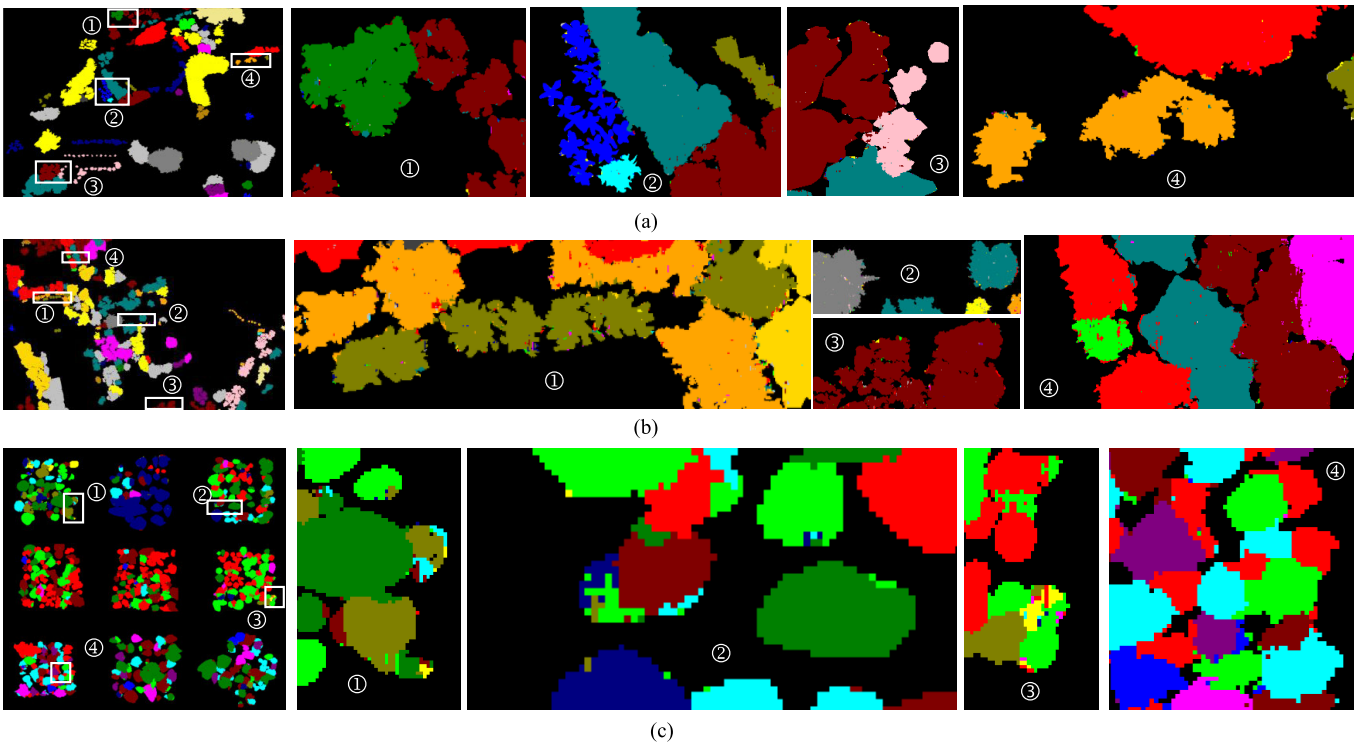


Fig. 16. Visualization of typical misclassification cases across three datasets: (a) SZUTreeData-R1. (b) SZUTreeData-R2. (c) MNFPTree-HL. The highlighted regions show representative failure patterns, primarily caused by canopy overlap, mixed pixels among similar species, and ambiguous boundaries, which increase feature confusion and lead to misclassification.

exhibits the highest visual fidelity to the ground truth. This visual evidence confirms our model's enhanced capability to synergistically fuse spectral and structural information, effectively disentangling complex canopies and validating its architectural design for real-world forestry applications.

D. Error Analysis

The normalized confusion matrices for SZUTreeData-R1, SZUTreeData-R2, and the natural secondary forest dataset demonstrate consistently strong classification performance. For SZUTreeData-R1, as shown in Fig. 15(a), the diagonal entries are close to 1, while off-diagonal values remain uniformly low (0–0.2), indicating that most tree species are correctly classified with negligible interclass confusion. No pronounced misclassification clusters are observed, suggesting effective feature separability across all categories. For SZUTreeData-R2, as shown in Fig. 15(b), a similar pattern is observed, with dominant diagonal responses and sparse off-diagonal distributions. Minor misclassifications mainly occur among morphologically and spectrally similar species, such as Ficus species (*Ficus macrophylla, Ficus concinna*, and *Ficus virens*), palm species (*Archontophoenix alexandrae* and *Livistona chinensis*), and Melaleuca with *Casuarina equisetifolia*, which can be attributed to their similar canopy structures and spectral characteristics. For the MNFPTree-HL dataset, as shown in Fig. 15(c), the confusion matrix exhibits near-perfect classification performance, with highly concentrated diagonal responses and negligible off-diagonal values. Some

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TABLE IV
ABLATION STUDY OF MODEL ARCHITECTURE ON SZUTREEDATA-R1

SSFE	GCCM	TADR	CLH	OA(%)	AA(%)	Kappa(%)
✗	✓	✓	✓	95.71	93.47	94.51
✓	✗	✓	✓	92.90	90.67	91.68
✓	✓	✗	✓	89.31	88.79	88.53
✓	✓	✓	✗	97.82	94.09	97.52
✓	✓	✓	✓	**98.36**	**95.47**	**98.13**

misclassifications are observed among species with similar ecological and structural traits, including Elm Tree and Shuiqu, Red Pine and Colored Wood, as well as Poplar Tree and Willow Tree.

Typical misclassified pixels for the three datasets are illustrated in Fig. 16. For SZUTreeData-R1 [Fig. 16(a)], misclassifications mainly occur in regions with severe canopy interlacing and overlap among closely related species (e.g., Ficus and Arecaceae), where mixed pixels integrate spectral and structural features from multiple species. In addition, boundary pixels with incomplete information further introduce ambiguity, leading to classification errors. For SZUTreeData-R2 [Fig. 16(b)], misclassification is more pronounced due to increased species diversity and stronger canopy mixing (e.g., Meliaceae, Myrtaceae, and Ficus). Overlapping canopy structures and slender crown morphologies reduce feature separability, while edge pixels with low distinguishability become the primary sources of errors. For the MNFPTree-HL dataset [Fig. 16(c)], dense mixed growth in natural secondary forests leads to significant canopy superposition. Complex background and ambiguous canopy boundaries further exacerbate feature mixing, resulting in boundary pixels being the main regions of misclassification. Overall, misclassifications across all datasets are primarily caused by canopy overlap and boundary ambiguity, which limit the model's ability to learn discriminative features.

TABLE V
COMPARISON OF TADR WITH SE AND AG MODULES ON SZUTREEDATA-R1

Module	OA (%)	AA (%)	Kappa (%)
SE	97.30	94.25	97.22
AG	97.50	94.58	97.08
TADR	**98.36**	**95.47**	**98.13**

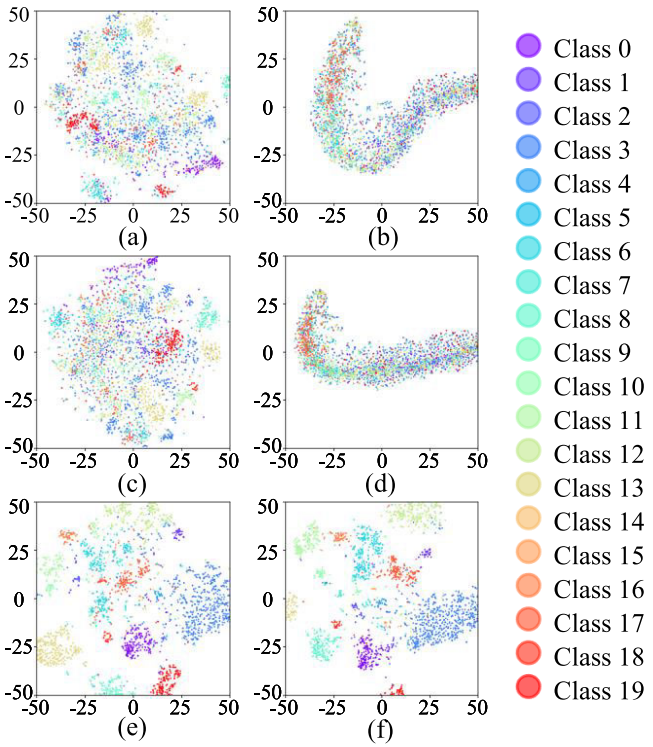


Fig. 17. t-SNE visualization results on SZUTreeData-R1 dataset. (a) Base feature from HSI. (b) Base feature from LiDAR. (c) Shallow feature fusion. (d) Local feature. (e) Global feature. (f) Deep–shallow feature fusion.

E. Ablation Studies

To validate the robustness of MS^{3}Mamba and quantify the contribution of its key design choices, we conducted six ablation experiments: 1) model architecture; 2) feature representation; 3) input data; 4) patch size; 5) number of hyperspectral bands in PCA; and 6) hyperparameter of loss function.

1) Model Architecture: To quantify the contribution of each key component, we conduct an ablation study on the SZUTreeData-R1 dataset by individually removing the SSFE, GCCM, TADR, and CLH modules (Table IV). The results reveal a clear hierarchy of importance. The most pronounced performance degradation occurs upon removing the TADR module (−8.97%), underscoring that the synergistic fusion of deep and shallow features is the most critical element of our architecture. This is followed by the ablation of the GCCM module (−5.38%), confirming the indispensable role of Mamba-based long-range contextual modeling. The removal of the SSFE (−2.57%) and CLH (−0.46%) modules also leads to notable decreases, validating their respective roles in initial feature extraction and final channel refinement. Collectively, this study quantitatively affirms that the hierarchical integration of global context and local detail is the central pillar of MS^{3}Mamba's success.

To investigate the boundary perception capability of different methods for tree species classification, we conducted a comparative experiment by replacing the proposed TADR module with the SE and AG modules, as summarized in Table V. The experimental results show that the proposed TADR module achieves the best performance across all three key metrics, with an OA of 98.36, an AA of 95.47, and a Kappa coefficient of 98.13, outperforming both the SE and AG modules. This improvement can be attributed to the tree-aware design of TADR, which enhances the perception of tree boundaries and enables more effective integration of global contextual information with local fine-grained details, thereby improving feature representation and classification accuracy. Overall, these results indicate that TADR provides a stronger capability for perceiving tree boundaries and capturing discriminative features than the compared methods.

2) Feature Representation: To further analyze how the architecture achieves this performance, we visualize the

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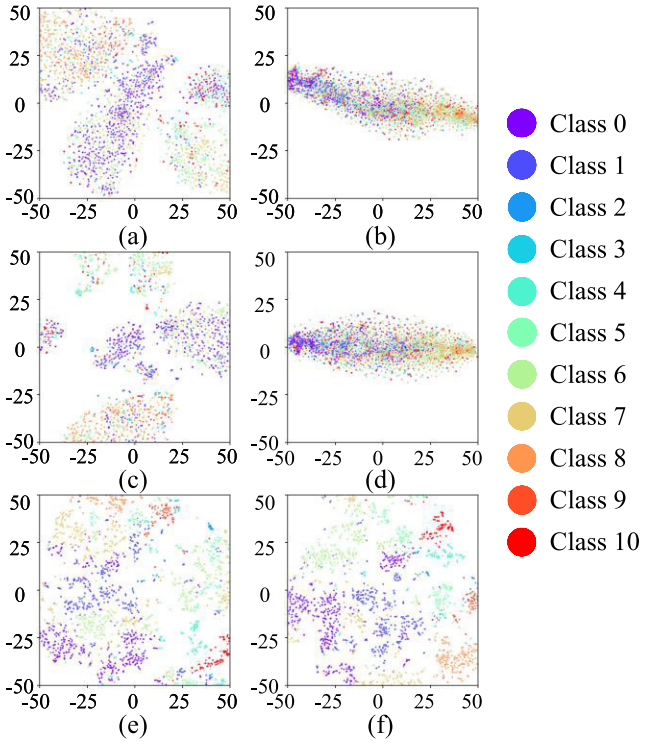


Fig. 18. t-SNE visualization results on MNFPTree-HL dataset. (a) Base feature from HSI. (b) Base feature from LiDAR. (c) Shallow feature fusion. (d) Local feature. (e) Global feature. (f) Deep–shallow feature fusion.

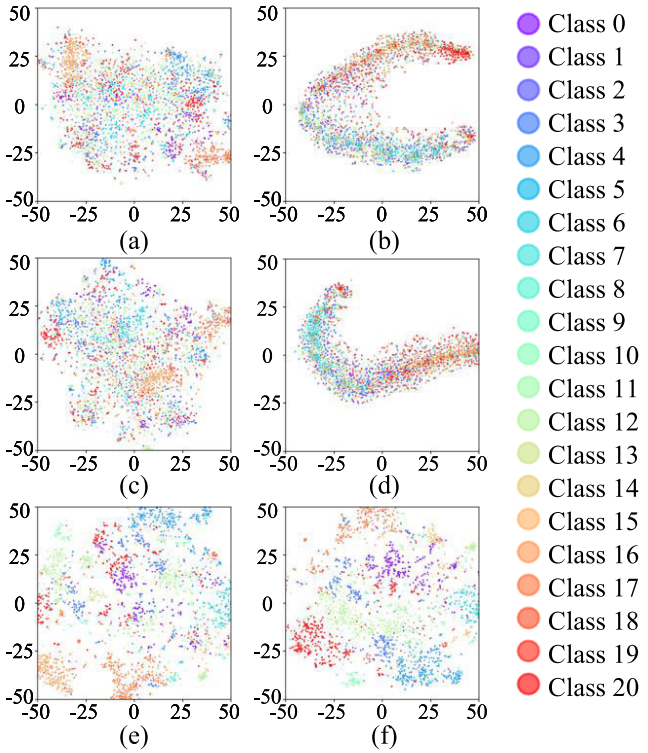


Fig. 19. t-SNE visualization results on SZUTreeData-R2 dataset. (a) Base feature from HSI. (b) Base feature from LiDAR. (c) Shallow feature fusion. (d) Local feature. (e) Global feature. (f) Deep–Shallow feature fusion.

TABLE VI
ABLATION STUDY WITH DIFFERENT INPUTS ON SZUTREEDATA-R1

HSI	LiDAR	OA(%)	AA(%)	Kappa(%)
✓	✗	97.05	95.06	97.38
✗	✓	49.19	29.59	41.13
✓	✓	**98.36**	**95.47**	**98.13**

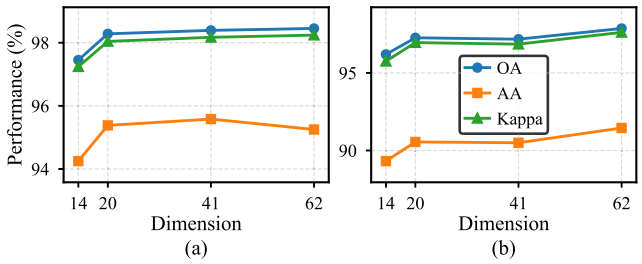


Fig. 20. Sensitivity analysis of PCA-based band reduction with different reduced dimensions on two datasets: (a) SZUTreeData-R1 and (b) SZUTreeData-R2. OA, AA, and Kappa are reported to evaluate the influence of dimensionality on classification performance.

feature distributions from different network layers using t-SNE (Figs. 17–19). The visualizations reveal a clear feature learning progression: while shallow-layer features exhibit significant class overlap, the deep-layer features progressively form compact and well-separated clusters, indicating high interclass separability and low intraclass variance. This progression qualitatively validates that our architecture effectively transforms the raw multimodal inputs into a highly discriminative feature space. Taken together with the quantitative ablation results, these visualizations confirm that MS^{3}Mamba learns a robust and generalizable representation by effectively leveraging complementary HSI and LiDAR information.

3) Input Data: To quantify the contribution of each data modality, we conduct an ablation study under three input configurations: LiDAR-only, HSI-only, and fused HSI-LiDAR (Table VI). The results unequivocally demonstrate the synergistic value of fusion. The LiDAR-only model performed poorly (49.19%), confirming that structural information alone is insufficient for this fine-grained task. Conversely, the HSI-only model achieves high accuracy (97.05%), establishing spectral information as the primary performance driver. Crucially, the fused model yields the highest accuracy (98.36%), an improvement of +1.23% over the HSI-only baseline. This improvement validates our architecture's ability to exploit LiDAR's complementary structural context to resolve ambiguities inherent in HSI, likely by aiding the delineation of overlapping crowns and the differentiation of spectrally similar species with distinct vertical profiles.

4) Patch Size: The input patch size, a pivotal hyperparameter governing the tradeoff between spatial context and computational cost, is optimized through an empirical analysis on the SZUTreeData-R1 dataset. As shown in Table VII, we evaluate patch sizes ranging from 7×7 to 15×15. The results indicate that classification accuracy generally improves with larger patches, increasing from 94.03% (7×7) to a peak of 98.64% (15×15). However, this trend exhibits diminishing returns, with the gain from 13×13 to 15×15 being marginal (+0.23%) at a substantial computational overhead.

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TABLE VII
ABLATION STUDY WITH PATCH SIZE ON SZUTREEDATA-R1

Patch Size	OA (%)	AA (%)	Kappa (%)	Params (M)	MACs (G)	FLOPs (G)	Peak Memory (GB)
7×7	94.03	89.47	93.21	0.556	13.16	26.32	0.24
9×9	97.02	93.68	96.60	0.558	21.79	43.59	0.38
11×11	98.36	95.47	98.13	0.561	32.53	65.07	0.57
13×13	98.41	95.31	98.19	0.564	45.49	90.98	0.79
15×15	98.64	95.71	98.45	0.567	60.53	121.06	1.04

TABLE VIII
COMPARISON OF CLASSIFICATION PERFORMANCE AND COMPUTATIONAL COST UNDER DIFFERENT PCA SPECTRAL DIMENSIONS ON SZUTREEDATA-R1 (1000 PIXELS)

Dimension	OA (%)	AA (%)	Kappa (%)	Params (M)	MACs (G)	FLOPs (G)	Peak Memory (GB)
14	97.45	94.25	97.24	0.51	28.39	56.79	0.56
20	98.36	95.47	98.13	0.56	32.53	65.06	0.57
41	98.39	95.58	98.17	0.75	47.95	95.89	0.80
62	98.45	95.52	98.24	0.98	63.99	127.98	1.03

TABLE IX
SENSITIVITY ANALYSIS OF LOSS FUNCTION HYPERPARAMETER ON SZUTREEDATA-R1

Hyperparameter	OA (%)	AA (%)	Kappa (%)
0.0	96.50	93.47	96.29
0.2	97.45	94.44	97.23
0.4	97.41	94.46	97.18
0.6	97.38	94.35	**98.15**
0.8	**98.36**	**95.47**	98.13
1.0	4.44	3.56	0.60

Therefore, to achieve a compelling balance between accuracy and efficiency, we select 11×11 as the optimal patch size for all subsequent experiments.

5) Number of Hyperspectral Bands in PCA: To investigate the effect of PCA dimensionality, sensitivity analyses were conducted on SZUTreeData-R1 and SZUTreeData-R2 using 14, 20, 41, and 62 principal components. These dimensionality settings were chosen by considering the model architecture, the progressive retention of spectral information after PCA, and the tradeoff between representation capability and computational cost. As shown in Fig. 20(a), for SZUTreeData-R1, OA and Kappa increase rapidly from 14 to 20 components and then remain stable around 0.983–0.985, indicating that sufficient discriminative information is preserved with 20 components. Although AA reaches its maximum at 41 components, the overall performance gain beyond 20 components is limited. For SZUTreeData-R2 [Fig. 20(b)], OA and Kappa improve steadily with increasing dimensionality, becoming relatively stable when the number of components is no less than 20, while AA also increases gradually from 0.893 to 0.914. In addition, as reported in Table VIII, the computational cost and peak memory usage increase continuously with the number of spectral components, whereas the performance improvement beyond 20 components is marginal. Therefore, 20 principal components were selected as a practical tradeoff between classification accuracy and computational efficiency.

6) Hyperparameter of Loss Function: To analyze the sensitivity of the loss hyperparameter η and validate its chosen value, we conducted a comparative experiment by varying η from 0.0 to 1.0 to evaluate its effect on the balance between feature loss and classification loss. The feature loss enhances interclass discriminability, while the classification loss supervises category prediction, and the two are complementary. As shown in Table IX, the model maintains relatively stable performance for η values from 0.0 to 0.8. However, when $\eta = 1.0$, the classification loss is completely removed, leading to a sharp performance drop. $\eta = 0.8$ achieves the best overall tradeoff, with the highest OA and AA, while $\eta = 0.6$ gives a slightly higher Kappa, demonstrating an effective balance between feature learning and classification optimization. When η is too small, insufficient feature constraints limit discriminability, while an excessively large η overemphasizes feature learning and weakens classification supervision, resulting in performance degradation, as evidenced by the sharp drop at $\eta = 1.0$. Therefore, $\eta = 0.8$ is chosen as a reasonable empirical choice.

F. Comparison of Calculation Costs

Beyond classification accuracy, the practical viability of a deep learning model is also determined by its computational efficiency. To evaluate this aspect, we conduct a comparative analysis of MS^{3}Mamba against SOTA methods in terms of model parameters (P), floating-point operations (FLOPs), training time (TR), and testing time (TE). The results, summarized across all three datasets in Table X, provide a comprehensive overview of the tradeoffs between model complexity and performance across different methods.

1) Training and Testing TEs: The results reveal a compelling tradeoff. The TR of MS^{3}Mamba (e.g., 136 min on SZUTreeData-R1), while longer than that of simpler CNN-based methods like CoupledCNN (50 min), is an expected consequence of its more sophisticated fusion architecture. Crucially, it remains substantially more efficient to train than other high performing, computationally intensive methods, such as NCGFL2 (508 min) and S3F2Net (987 min). Moreover, its inference TE (e.g., 111 s on SZUTreeData-R1) is highly competitive and comparable to most advanced methods, confirming its practicality for operational deployment.

2) Model Complexity: MS^{3}Mamba is engineered to strike an effective balance between model complexity and classifica-

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TABLE X
COMPARISON OF TR, TESTING TE, PARAMETERS (P), AND FLOPS (F) ON THREE DATASETS. FLOPS DENOTED THE COMPUTATIONAL COST PER PIXEL

Methods	SZUTreeData-R1				SZUTreeData-R2				MNFPTree-HL			
	TR(min)	TE(s)	P(K)	F(M)	TR(min)	TE(s)	P(K)	F(M)	TR(s)	TE(s)	P(K)	F(M)
CoupledCNN [16]	50	203	107	5	66	136	107	5	74	6	106	5
MAHiDFNet [44]	83	118	45280	482	113	165	45283	482	325	13	45275	482
HCTnet [36]	53	109	407	12	72	121	407	12	112	12	406	12
CALC [17]	61	120	798	23	72	281	799	23	565	3	795	23
MFT [19]	55	90	259	17	72	115	259	17	77	4	257	17
MVit [20]	59	87	159	51	77	110	159	51	106	4	158	51
NCGFL [40]	508	592	521	163	721	782	522	163	1655	30	521	163
HLMamba [23]	42	101	456	202	59	143	456	202	168	6	454	202
IDnet [22]	168	408	148	66	139	229	148	66	472	11	148	66
S3F2Net [21]	987	1118	281	993	1889	2118	281	1041	5	2	279	102
MS^{3}Mamba (proposed)	136	111	308	65	186	147	307	65	283	4	306	65

tion performance. With approximately 308 K parameters and 65 M FLOPs, it is substantially more parsimonious than large attention-based models (e.g., MAHiDFNet, 45.28 M) and is orders of magnitude more efficient than graph-based methods (e.g., S3F2Net, 97.3 G). Although lightweight CNN-based models are more compact (e.g., CoupledCNN, 107 K parameters), MS^{3}Mamba achieves top-tier accuracy with superior parameter efficiency, avoiding the parameter growth common in other advanced fusion networks. This efficiency extends to its practical TE costs (Table X); although its advanced fusion architecture requires more TR than basic CNN, the model remains computationally tractable and maintains a competitive inference speed. This compelling tradeoff between accuracy and computational overhead confirms the viability of MS^{3}Mamba as a practical and powerful solution for real-world applications.

VI. CONCLUSION

In this study, we proposed MS^{3}Mamba, a novel end-to-end architecture for fine-grained tree species classification that synergistically fuses HSI and LiDAR data. Its efficacy is rooted in a hierarchical fusion strategy that systematically reconciles global context with local detail. This is achieved through a dual-stream spatial–spectral Mamba for efficient long-range dependency modeling, complemented by a deep–shallow fusion mechanism that integrates deep contextual representations with shallow, high-frequency details. This integrated design effectively mitigates the challenges of high intraspecies variability and interspecies similarity, particularly in delineating overlapping canopies. Comprehensive experiments across three datasets, including our newly established and challenging MNFPTree-HL dataset from a natural secondary forest, unequivocally validate the robustness and superiority of MS^{3}Mamba. Our architecture demonstrated superior performance over a wide range of SOTA methods, confirming it as a powerful and efficient framework for tree species classification demanding the simultaneous reconciliation of contextual patterns and fine-grained boundary details.

Future work will be pursued along two main directions. First, we plan to expand the dataset and adopt spatially disjoint splits (i.e., using independent geographic regions for training and testing) to further verify the generalization ability of the model. Second, we aim to further improve tree species classification accuracy in complex stands by refining canopy boundary extraction and enhancing feature learning in overlapping regions, thereby reducing misclassification.

DATA AVAILABILITY

The MNFPTree-HL dataset is available from the corresponding author upon reasonable request.

REFERENCES

[1] Z. Zhou, J. Wu, C. Wang, J. Wang, and F. Cheng, "A synchronous acquisition method for dominant tree species and forest age in complex mountainous terrain through growth characteristics matching," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4403518.

[2] Y. Long, S. Ye, L. Wang, W. Wang, X. Liao, and S. Jia, "Scale pyramid graph network for hyperspectral individual tree segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5526014.

[3] F. E. Fassnacht et al., "Review of studies on tree species classification from remotely sensed data," *Remote Sens. Environ.*, vol. 186, pp. 64–87, Dec. 2016.

[4] N. Li, S. Jiang, J. Xue, S. Ye, and S. Jia, "Texture-aware self-attention model for hyperspectral tree species classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502215.

[5] L. Chen, Y. Wei, Z. Yao, E. Chen, and X. Zhang, "Data augmentation in prototypical networks for forest tree species classification using airborne hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410116.

[6] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[7] F. Tong and Y. Zhang, "Spectral–spatial and cascaded multilayer random forests for tree species classification in airborne hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4411711.

[8] M. Ahmad et al., "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[9] X. Shen and L. Cao, "Tree-species classification in subtropical forests using airborne hyperspectral and LiDAR data," *Remote Sens.*, vol. 9, no. 11, p. 1180, Nov. 2017.

[10] H. Zhong et al., "Identification of tree species based on the fusion of UAV hyperspectral image and LiDAR data in a coniferous and broad-leaved mixed forest in northeast China," *Frontiers Plant Sci.*, vol. 13, Sep. 2022, Art. no. 964769.

[11] D. Zhao, Y. Pang, L. Liu, and Z. Li, "Individual tree classification using airborne LiDAR and hyperspectral data in a natural mixed forest of northeast China," *Forests*, vol. 11, no. 3, p. 303, Mar. 2020.

[12] Y. Quan, M. Li, Y. Hao, J. Liu, and B. Wang, "Tree species classification in a typical natural secondary forest using UAV-borne LiDAR and hyperspectral data," *GIScience Remote Sens.*, vol. 60, no. 1, Dec. 2023, Art. no. 2171706.

[13] B. Wang, J. Liu, J. Li, and M. Li, "UAV LiDAR and hyperspectral data synergy for tree species classification in the Maoershan forest farm region," *Remote Sens.*, vol. 15, no. 4, p. 1000, Feb. 2023.

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[14] J. Mäyrä et al., "Tree species classification from airborne hyperspectral and LiDAR data using 3D convolutional neural networks," *Remote Sens. Environ.*, vol. 256, Apr. 2021, Art. no. 112322.
[15] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[16] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.
[17] T. Lu, K. Ding, W. Fu, S. Li, and A. Guo, "Coupled adversarial learning for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 93, pp. 118–131, May 2023.
[18] Y. Ma, Y. Zhao, J. Im, Y. Zhao, and Z. Zhen, "A deep-learning-based tree species classification for natural secondary forests using unmanned aerial vehicle hyperspectral images and LiDAR," *Ecological Indicators*, vol. 159, Feb. 2024, Art. no. 111608.
[19] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.
[20] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.
[21] X. Wang, L. Song, Y. Feng, and J. Zhu, "S3F2Net: Spatial–spectral-structural feature fusion network for hyperspectral image and LiDAR data classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4801–4815, May 2025.
[22] C. Liang, Y. Zhao, Y. Song, and K. Ni, "IDNet: Intensity-constrained detail-enhanced network for hyperspectral and LiDAR collaborative classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5518515.
[23] D. Liao, Q. Wang, T. Lai, and H. Huang, "Joint classification of hyperspectral and LiDAR data base on mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530915.
[24] H. Liu and C. Wu, "Crown-level tree species classification from AISA hyperspectral imagery using an innovative pixel-weighting approach," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 68, pp. 298–307, Jun. 2018.
[25] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[26] Y. Xu, B. Du, and L. Zhang, "Beyond the patchwise classification: Spectral–spatial fully convolutional networks for hyperspectral image classification," *IEEE Trans. Big Data*, vol. 6, no. 3, pp. 492–506, Sep. 2020.
[27] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.
[28] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.
[29] J. Wu, L. Chen, J. Wang, Y. Li, E. Chen, and X. Zhang, "A novel framework combining band selection algorithm and improved 3D prototypical network for tree species classification using airborne hyperspectral images," *Comput. Electron. Agricult.*, vol. 219, Apr. 2024, Art. no. 108813.
[30] H. Oshio, T. Asawa, A. Hoyano, and S. Miyasaka, "Estimation of the leaf area density distribution of individual trees using high-resolution and multi-return airborne LiDAR data," *Remote Sens. Environ.*, vol. 166, pp. 116–125, Sep. 2015.
[31] M. Kukkonen, T. Lähivaara, and P. Packalen, "Combination of LiDAR intensity and texture features enable accurate prediction of common boreal tree species with single sensor UAS data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4401508.
[32] Y. Liu, Y. Chen, Z. Liu, J. Chen, and Y. Liu, "A multifeature fusion network for tree species classification based on ground-based LiDAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4648–4663, 2025.
[33] Y. Ma, P. Luan, Y. Zhang, B. Liu, and L. Zhang, "Efficient urban tree species classification via multirepresentation fusion of mobile laser scanning data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 11451–11468, 2025.
[34] P. Zhang, T. Li, G. Wang, D. Wang, P. Lai, and F. Zhang, "A multi-source information fusion model for outlier detection," *Inf. Fusion*, vol. 93, pp. 192–208, May 2023.
[35] Y. Zhou et al., "Spatiotemporal fusion of multi-source remote sensing data for estimating aboveground biomass of grassland," *Ecol. Indicators*, vol. 146, Feb. 2023, Art. no. 109892.
[36] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.
[37] T. Sankey, J. Donager, J. McVay, and J. B. Sankey, "UAV LiDAR and hyperspectral fusion for forest monitoring in the southwestern USA," *Remote Sens. Environ.*, vol. 195, pp. 30–43, Jun. 2017.
[38] C. Sothe et al., "Comparative performance of convolutional neural network, weighted and conventional support vector machine and random forest for classifying tree species using hyperspectral and photogrammetric data," *GIScience Remote Sens.*, vol. 57, no. 3, pp. 369–394, Apr. 2020.
[39] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
[40] B. Tu, Q. Ren, J. Li, Z. Cao, Y. Chen, and A. Plaza, "NCGLF2: Network combining global and local features for fusion of multisource remote sensing data," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102192.
[41] X. Li et al., "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. Adv. Neural Inf. Process. Systems. (NeurIPS)*, 2020, pp. 21002–21012.
[42] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024.
[43] S. Zhang, M. Xu, J. Zhou, and S. Jia, "Unsupervised spatial–spectral CNN-based feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524617.
[44] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

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